A sharp three-window threshold and finite-memory conjugacy in Fibonacci stabilization

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Abstract

We study the canonical Fibonacci stabilization map

$$\mathrm{Fold}_m : \{0,1\}^m \to X_m,$$

which sends a binary word to the unique admissible Fibonacci word with the same valuation modulo F_{m+2}. The local map is many-to-one on isolated windows, but the relevant question is whether this ambiguity survives under sliding overlap. We prove a sharp threshold at $m = 3$. For every $m \geq 3$, the overlapping stabilized-window map is injective, admits an explicit finite-memory inverse, and identifies its image with a subshift of finite type topologically conjugate to the full two-shift. At $m = 2$, the failure of injectivity is completely concentrated on a unique branch orbit; this yields exact formulas for the conditional entropy and Kullback–Leibler defect and implies mod-zero invertibility off the branch set. On the arithmetic side, stabilization fibers admit an exact residue characterization and a discrete Fourier formula, and their second moment satisfies a linear recurrence. On path space, the conjugacy transports invariant measures, entropy, continuous potentials, pressure, and equilibrium states exactly. In the Sturmian benchmark this also yields deterministic discrepancy bounds for stabilized block laws and for the recovery of finite-range observables. The cut-and-project input is used only to fix the source law and admissible grammar; the theorem-level core is symbolic and arithmetic.

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1 Introduction

Finite-window recodings arise naturally in symbolic dynamics, numeration, and finite-resolution observation. A local map may be many-to-one on isolated words and yet become invertible once one passes to overlapping windows along a trajectory. The problem studied

here is whether this recovery-by-overlap occurs for the canonical Fibonacci stabilization map, and if so at what window size.

For each $m \geq 1$ we consider

$$\mathrm{Fold}_m : \{0,1\}^m \to X_m,$$

the map that sends a binary word to the unique Fibonacci-admissible word with the same valuation modulo F_{m+2}. On individual words Fold_m is genuinely many-to-one: different binary blocks can lie in the same residue class and therefore stabilize to the same admissible block. The central question is whether this local folding persists on path space for the associated sliding-block map, or whether overlap restores the lost information.

The answer is sharp. The main contributions of the paper are:

(i) The overlap graph yields a finite-state lift with bounded degree, and the threshold window is exactly $m = 3$: Lemma 4.2 gives an explicit three-window decoder and Theorem 4.3 proves injectivity for every $m \geq 3$.

(ii) The exceptional case $m = 2$ is completely rigid rather than generically non-injective. Theorem 4.6 identifies a unique branch orbit, Corollary 4.7 gives exact entropy and Kullback–Leibler defect formulas, and Corollary 4.9 shows mod-zero invertibility off that branch set.

(iii) Above threshold, the stabilized image is a finite-type recoding of the full two-shift. Theorem 4.12 supplies a finite-memory inverse and identifies the image shift Y_m as a subshift of finite type.

(iv) The conjugacy has exact dynamical consequences. Proposition 4.13 and Theorem 6.4 transport invariant measures, entropy, continuous potentials, pressure, and equilibrium states exactly, while Section 8 gives quantitative discrepancy-based corollaries in the Sturmian benchmark.

This places the paper closer to symbolic dynamics and arithmetic coding than to a measurement-theory manuscript. The cut-and-project and detector language serve only to fix the benchmark source law and to explain why the local problem must be posed on the full cube $\{0,1\}^m$. Once the source law and admissible grammar are fixed, the proofs are symbolic and arithmetic.

The relevant literature splits into four strands. Classical work on Sturmian words and Fibonacci numeration explains admissibility, normalization, and automaton structure at the level of words [16, 20, 24, 18, 19]; the broader interaction between numeration systems and symbolic dynamics is surveyed in [35], and S-adic generalizations appear in [38]. General sofic-factor theory provides the language of labeled graphs, resolving maps, and finite-to-one factors [17, 13, 14]; within that framework, resolving factor maps between irreducible shifts of finite type with equal entropy are known to be conjugacies under general conditions [36], and lower-entropy sofic factors have been characterized abstractly [37]. The connection between β-expansions and shift spaces was introduced by Parry [34]. Standard thermodynamic formalism explains what must be preserved under topological conjugacy [29]. What these frameworks do not provide on their own is the arithmetic threshold phenomenon proved here: the explicit fact that $m = 2$ fails in exactly one branch locus, whereas every $m \geq 3$ admits a finite-memory inverse. That conclusion depends on the residue arithmetic of Fibonacci stabilization and, in particular, on the decoder of Lemma 4.2; it is not a formal consequence of the general sofic setup.

To summarize the new content explicitly: the sharp $m = 3$ threshold (Theorem 4.3), together with the constructive decoder that proves it (Lemma 4.2); the complete branch-locus classification at $m = 2$ (Theorem 4.6) with exact entropy and Kullback–Leibler defect formulas (Corollary 4.7); the finite-memory conjugacy to the full two-shift (Theorem 4.12); the discrete Fourier formula for fiber cardinalities (Theorem 3.4); the second-moment linear recurrence (Theorem 7.1); and the affine transfer principle connecting canonical and shifted-weight residue models (Theorem 3.6). The thermodynamic transport theorems of Section 6 follow from the conjugacy by standard machinery, but their explicit local form in this arithmetic setting has not appeared before.

A second conceptual point is that fixed-window aliasing and path-space information loss are different objects. The entropy ledger of Section 5 measures the unresolved multiplicity of isolated windows. Above threshold this does not become a new source of dynamical irreversibility, because Section 6 proves that overlap reconstructs the raw trajectory exactly. The local defect at $m = 2$ is therefore a sharp exceptional branch, not evidence of a generic extensive loss mechanism.

The Fibonacci benchmark is a natural place to study this question. It is the smallest aperiodic binary system in which the admissible language, residue fibers, threshold decoder, and discrepancy bounds can all be written explicitly. This makes it possible to prove the path-space consequences at theorem level rather than to suggest them heuristically.

The original motivation nevertheless comes from finite-resolution observation of quasiperiodic order [1, 2, 7]. In the Sturmian benchmark, any strictly positive local detector noise makes every binary word of length m appear with positive probability, so the ambient cube is physically forced at the readout layer. That application explains why the local folding problem is worth studying, but it does not determine the mathematical identity of the paper.

The paper is organized as follows. Section 2 records the symbolic setup, the Sturmian benchmark, and the full-support mechanism. Section 3 defines Fibonacci stabilization on words, gives the residue description of the fibers, and includes a concrete worked example. Section 4 develops the overlap graph, the sharp three-window threshold, and the two-window branch locus. Section 5 isolates the fixed-window aliasing ledger, and Section 6 turns the threshold theorem into finite-memory conjugacy, path-space equivalence, and transport of thermodynamic formalism. Sections 7 and 8 record arithmetic fallout and quantitative discrepancy bounds in the Sturmian benchmark. Appendix A contains the source-layer model-set input.

2 Symbolic setup and support inflation

The theorem-level results concern a local Fibonacci folding on the full cube and its overlap dynamics. This section records only the symbolic input imported from the source layer: the Sturmian benchmark fixes the admissible grammar, and strictly positive local noise explains why the local problem must be posed on all of $\{0, 1\}^m$ rather than only on admissible words.

2.1 Coding map and induced law

At the symbolic layer we only need a measurable coding map from the source dynamics into a finite alphabet. The detailed detector parameters are suppressed here; they matter only through the induced symbolic law.

Definition 2.1 (Finite-alphabet readout)**.** Fix a measurable map $R : H \to \mathcal{A}$ into a finite alphabet $\mathcal{A}$. Given a scan sequence $(h_t)_{t\in\mathbb{N}} \subset H$ (typically $h_t = h_0 + t\alpha$), the *readout process* is the symbolic sequence

$$(A_t)_{t\in\mathbb{N}}, \qquad A_t := R(h_t) \in \mathcal{A}.$$

In the simplest case $\mathcal{A} = \{0, 1\}$ and $R(h) = \mathbf{1}_W(h)$, this is a binary indicator readout. Finite-resolution variants replace the sharp indicator by a smoothed kernel that converges to $\mathbf{1}_W$ as the resolution parameter tends to zero.

Definition 2.2 (Coding map and induced input law)**.** Let $(H, \mathcal{B}, \mu, T)$ be the scan dynamics at the source layer, where μ is T-invariant. The *coding map* induced by R is

$$\iota_R : H \to \mathcal{A}^{\mathbb{N}}, \qquad \iota_R(h) := \Big(R(T^t h)\Big)_{t\geq 0}.$$

Let $\sigma : \mathcal{A}^{\mathbb{N}} \to \mathcal{A}^{\mathbb{N}}$ denote the left shift. The *induced input law* on symbolic sequences is the pushforward

$$\nu_R := (\iota_R)_*\mu.$$

Proposition 2.1 (Stationarity and ergodicity of the induced law)**.** *The induced law ν_R is σ-invariant. If μ is ergodic for T, then ν_R is ergodic for σ.*

Proof. For a cylinder $[a_0 \cdots a_{m-1}]$ in $\mathcal{A}^{\mathbb{N}}$,

$$\nu_R(\sigma^{-1}[a_0 \cdots a_{m-1}]) = \mu\Big(\{h : \ R(T^{t+1}h) = a_t, \ t = 0, \ldots, m-1\}\Big).$$

By T-invariance of μ, this equals

$$\mu\Big(\{h : \ R(T^t h) = a_t, \ t = 0, \ldots, m-1\}\Big) = \nu_R([a_0 \cdots a_{m-1}]),$$

so ν_R is shift-invariant. Ergodicity is preserved under factors [29]. □

2.2 The Fibonacci/Sturmian benchmark

When the scan dynamics is an irrational circle rotation and the readout is a two-interval coding, the resulting symbolic process is Sturmian [16, 20, 24]. This benchmark is singled out by its minimal complexity among aperiodic binary codings and by the fact that, in the golden regime, it lands naturally in the grammar used later for stabilization.

Proposition 2.2 (Rotation-to-grammar bridge in the golden regime)**.** *Let $T : \mathbb{T}^1 \to \mathbb{T}^1$ be the rotation $T(\theta) = \theta + \alpha \pmod 1$ and take the readout*

$$R(\theta) := \mathbf{1}_{[0,\,1-\alpha)}(\theta).$$

If $\alpha \geq \frac{1}{2}$, then the induced symbolic process $A_t := R(T^t\theta_0)$ is 11-free, hence belongs to the binary subshift that forbids the word 11. In particular this holds for the golden slope $\alpha = \varphi^{-1}$.

Proof. If $A_t = 1$, then $\theta_t := T^t\theta_0 \in [0, 1-\alpha)$, so $\theta_{t+1} = \theta_t + \alpha \in [\alpha, 1)$. When $\alpha \geq \frac{1}{2}$ the intervals $[0, 1-\alpha)$ and $[\alpha, 1)$ are disjoint, so $A_{t+1} = 0$. □

Remark 2.1 (Role of the Sturmian benchmark). Proposition 2.2 shows that a noiseless Fibonacci source lands inside the admissible language X_m. Combined with Theorem 3.2(ii), this means that local stabilization is trivial on the benchmark source itself: Fold_m acts as the identity on every observed word. The benchmark is retained because it canonically supplies the grammar used by stabilization and because its linear word complexity drives the total-variation bound of Section 8.

2.3 Support inflation to the full cube

The noiseless Fibonacci benchmark itself does not populate the ambient cube. The next theorem explains why the local stabilization problem must nevertheless be posed on all of $\{0,1\}^m$ once the readout layer contains any strictly positive local uncertainty.

Theorem 2.3 (Non-degenerate local noise fills the ambient cube)**.** *Let $B = (B_t)_{t\in\mathbb{N}}$ be the ideal Fibonacci/Sturmian benchmark process from Proposition 2.2, taken under its stationary rotation law. Let $A = (A_t)_{t\in\mathbb{N}}$ be obtained from B through a strictly positive local binary channel $Q = (Q_{ba})_{a,b\in\{0,1\}}$ in the sense that, conditional on any raw block $b_0 \cdots b_{m-1}$,*

$$\mathbb{P}(A_0 \cdots A_{m-1} = a_0 \cdots a_{m-1} \mid B_0 \cdots B_{m-1} = b_0 \cdots b_{m-1}) = \prod_{t=0}^{m-1} Q_{b_t a_t},$$

and assume $Q_{ba} > 0$ for all $a, b \in \{0,1\}$. Then for every $m \geq 1$ and every word $\omega \in \{0,1\}^m$,

$$\mathbb{P}(A_0 \cdots A_{m-1} = \omega) > 0.$$

In particular, the observable support of the noisy length-m measurement law is the full ambient cube $\{0,1\}^m$.

Proof. Fix $m \geq 1$ and $\omega \in \{0,1\}^m$. The depth-m Sturmian partition from Section 2.4 has nonempty atoms, so there exists at least one word $x = x_0 \cdots x_{m-1}$ in the support of $B_0 \cdots B_{m-1}$ with

$$\mathbb{P}(B_0 \cdots B_{m-1} = x) > 0.$$

Conditioning on that event gives

$$\mathbb{P}(A_0 \cdots A_{m-1} = \omega) \geq \mathbb{P}(B_0 \cdots B_{m-1} = x) \prod_{t=0}^{m-1} Q_{x_t \omega_t},$$

and the right-hand side is strictly positive because every channel entry is positive. □

Proposition 2.4 (Detector realization of strict positivity)**.** *Let $B = (B_t)_{t\in\mathbb{N}}$ be the ideal Fibonacci/Sturmian benchmark process. Fix an interaction radius $L \geq 0$, coefficients $\rho_{-L}, \ldots, \rho_L \geq 0$ with $\sum_{j=-L}^{L} \rho_j > 0$, a threshold $\tau \in \mathbb{R}$, and i.i.d. detector noises $(\xi_t)_{t\in\mathbb{N}}$ independent of B, whose common law has a density that is strictly positive on $\mathbb{R}$. Define the finite-resolution detector signal and binary output by*

$$S_t := \sum_{j=-L}^{L} \rho_j B_{t+j}, \qquad A_t := \mathbf{1}_{\{S_t + \xi_t \geq \tau\}}.$$

Then, conditional on any neighborhood block $b_{-L} \cdots b_{m-1+L} \in \{0,1\}^{m+2L}$, the output block law factorizes as

$$\begin{aligned}\mathbb{P}(A_0 \cdots A_{m-1} = a_0 \cdots a_{m-1} \mid B_{-L} \cdots B_{m-1+L} = b_{-L} \cdots b_{m-1+L}) \\ = \prod_{t=0}^{m-1} p_t(b)^{a_t} (1 - p_t(b))^{1-a_t}.\end{aligned}$$

for some probabilities $p_t(b) \in (0,1)$. In particular the detector induces a strictly positive local binary channel in the sense of Theorem 2.3, so every word $\omega \in \{0,1\}^m$ has positive probability under the measured output process.

Proof. Fix a neighborhood block $b := b_{-L} \cdots b_{m-1+L}$. For each $t = 0, \ldots, m-1$, the analog signal

$$s_t(b) := \sum_{j=-L}^{L} \rho_j b_{t+j}$$

is a deterministic real number. Since the common noise law of ξ_t has a density that is strictly positive on $\mathbb{R}$, the tail probability

$$p_t(b) := \mathbb{P}(\xi_t \geq \tau - s_t(b))$$

lies strictly between 0 and 1. Independence of the detector noises therefore gives the displayed product formula.

Now fix $\omega \in \{0,1\}^m$. Choose any neighborhood block b with

$$\mathbb{P}(B_{-L} \cdots B_{m-1+L} = b) > 0;$$

such a block exists because the support of the benchmark process is nonempty. Conditioning on that event and using the product formula yields

$$\mathbb{P}(A_0 \cdots A_{m-1} = \omega) \geq \mathbb{P}(B_{-L} \cdots B_{m-1+L} = b) \prod_{t=0}^{m-1} p_t(b)^{\omega_t}(1 - p_t(b))^{1-\omega_t} > 0.$$

Thus the detector realization induces a strictly positive local binary channel and fills the ambient cube. □

Remark 2.2 (Why the ambient extension is physically forced). Theorem 2.3 upgrades the motivational role of the ambient binary extension to an observable-support statement, and Proposition 2.4 realizes its hypothesis with a concrete finite-resolution detector. Once the readout layer has nonzero local uncertainty, the experimental length-m support leaves the Sturmian language and fills the full cube. The stabilization problem is therefore naturally posed on $\{0,1\}^m$ in the presence of finite-resolution readout noise.

2.4 Empirical block laws for the benchmark

For each depth m, the induced law ν_R determines a raw-word marginal on $\{0,1\}^m$. In the irrational-rotation benchmark, these marginals arise from the partition of $\mathbb{T}^1$ into nonempty depth-m Sturmian atoms, whose number is at most $m+1$. This linear complexity is the only source-specific input needed in Section 8.

3 Fibonacci stabilization on words

This section fixes the local arithmetic object of the paper. Stabilization is defined by reducing Fibonacci valuations modulo F_{m+2} and returning the unique admissible representative. The resulting map Fold_m is genuinely many-to-one on isolated words, and its residue fibers are the arithmetic input for the threshold and branch-locus theorems proved later.

3.1 Admissible language and Fibonacci valuation

Let $(F_n)_{n\geq 1}$ be the Fibonacci sequence with $F_1 = F_2 = 1$ and $F_{n+2} = F_{n+1} + F_n$.

Definition 3.1 (Golden-mean legal language)**.** For $m \geq 1$, let

$$X_m := \Big\{ x = (x_1, \ldots, x_m) \in \{0,1\}^m : x_i x_{i+1} = 0 \text{ for all } 1 \leq i < m \Big\}.$$

Definition 3.2 (Fibonacci valuation)**.** For $\omega = (\omega_1, \ldots, \omega_m) \in \{0,1\}^m$, define

$$\mathrm{Val}_m(\omega) := \sum_{k=1}^{m} \omega_k F_{k+1}.$$

Proposition 3.1 (Finite-window Zeckendorf range)**.** *For every $m \geq 1$, the restriction*

$$\mathrm{Val}_m|_{X_m} : X_m \longrightarrow \{0, 1, \ldots, F_{m+2} - 1\}$$

is a bijection.

Proof. By Zeckendorf's theorem, every integer $N \geq 0$ admits a unique expansion

$$N = \sum_{k \geq 1} \varepsilon_k F_{k+1}, \qquad \varepsilon_k \in \{0,1\}, \qquad \varepsilon_k \varepsilon_{k+1} = 0$$

[26, 20]. If $0 \leq N < F_{m+2}$, then the largest Fibonacci weight that can appear is at most F_{m+1}, so the Zeckendorf expansion of N has length at most m and defines a unique element of X_m. Thus $\mathrm{Val}_m|_{X_m}$ is surjective.

To prove injectivity, let $x, y \in X_m$ with $\mathrm{Val}_m(x) = \mathrm{Val}_m(y)$. Then x and y are two Zeckendorf expansions of the same integer using weights up to F_{m+1}, hence $x = y$ by uniqueness.

Equivalently, admissible words of length m realize each residue class $0, 1, \ldots, F_{m+2} - 1$ exactly once. □

Remark 3.1. Proposition 3.1 is the finite-window form of Zeckendorf uniqueness that makes a canonical stabilization map possible without any ambiguity at the right boundary. In that sense it is the rigid finite-window analogue of the finite-automaton normalization viewpoint developed for Fibonacci numeration in [18, 19].

3.2 Canonical finite-window stabilization

Definition 3.3 (Canonical stabilization map)**.** For $m \geq 1$, define

$$\mathrm{Fold}_m : \{0,1\}^m \longrightarrow X_m$$

by the rule

$$\mathrm{Fold}_m(\omega) := \Big(\mathrm{Val}_m|_{X_m}\Big)^{-1}\Big(\mathrm{Val}_m(\omega) \pmod{F_{m+2}}\Big),$$

where the residue is taken in $\{0, 1, \ldots, F_{m+2} - 1\}$.

Remark 3.2. The map Fold_m is the unique finite-window canonicalization compatible with Fibonacci numeration: compute the raw Fibonacci-weighted value $\mathrm{Val}_m(\omega)$, reduce it modulo F_{m+2}, and write the residue in its unique length-m Zeckendorf form. Local carry rewrites such as $11 \mapsto 100$ are one implementation of this residue-class stabilization.

Theorem 3.2 (Basic properties of Fold_m)**.** *For every $m \geq 1$, the stabilization map $\mathrm{Fold}_m : \{0,1\}^m \to X_m$ satisfies:*

(i) ***Well-definedness.*** Fold_m *is well-defined.*

(ii) ***Idempotence on admissible words.*** *If* $x \in X_m$*, then* $\mathrm{Fold}_m(x) = x$*.*

(iii) ***Surjectivity.*** Fold_m *is onto* X_m*.*

(iv) ***Fiber partition.*** *The fibers*

$$\mathcal{F}_m(x) := \mathrm{Fold}_m^{-1}(\{x\}), \qquad x \in X_m,$$

form a partition of $\{0,1\}^m$*. In particular,*

$$\sum_{x \in X_m} d_m(x) = 2^m, \qquad d_m(x) := |\mathcal{F}_m(x)|.$$

Proof. (i) follows from Proposition 3.1, which gives a unique admissible representative for each residue class modulo F_{m+2}.

(ii) If $x \in X_m$, then $\mathrm{Val}_m(x) \in \{0, \dots, F_{m+2} - 1\}$ and $\left(\mathrm{Val}_m|_{X_m}\right)^{-1}(\mathrm{Val}_m(x)) = x$.

(iii) Every $x \in X_m$ is a fixed point of Fold_m, so x lies in the image.

(iv) Since Fold_m is a function on the finite set $\{0,1\}^m$, its inverse images form a partition. Summing the cardinalities of the fibers gives the cardinality of the domain. □

3.3 Residue fibers and Fourier formula

Theorem 3.3 (Exact residue characterization of fibers)**.** *Let* $x \in X_m$ *and write*

$$r_x := \mathrm{Val}_m(x) \in \{0, \dots, F_{m+2} - 1\}.$$

Then

$$\mathcal{F}_m(x) = \Big\{\omega \in \{0,1\}^m : \mathrm{Val}_m(\omega) \equiv r_x \pmod{F_{m+2}}\Big\}.$$

Equivalently,

$$d_m(x) = \#\Big\{\omega \in \{0,1\}^m : \sum_{k=1}^{m} \omega_k F_{k+1} \equiv r_x \pmod{F_{m+2}}\Big\}.$$

Proof. By definition,

$$\mathrm{Fold}_m(\omega) = x \iff \left(\mathrm{Val}_m|_{X_m}\right)^{-1}\Big(\mathrm{Val}_m(\omega) \pmod{F_{m+2}}\Big) = x.$$

Since $\mathrm{Val}_m|_{X_m}$ is bijective, this is equivalent to

$$\mathrm{Val}_m(\omega) \equiv \mathrm{Val}_m(x) \pmod{F_{m+2}}.$$

This proves the claim. □

Remark 3.3 (Worked example at $m = 5$). At $m = 5$ the admissible language X_5 has $F_7 = 13$ words. The raw word $\omega = 11010$ has valuation

$$\mathrm{Val}_5(11010) = F_2 + F_3 + F_5 = 1 + 2 + 5 = 8.$$

The unique admissible word of length five with residue 8 modulo 13 is 00001, since $\mathrm{Val}_5(00001) = F_6 = 8$. Thus

$$\mathrm{Fold}_5(11010) = 00001.$$

In fact

$$\mathcal{F}_5(00001) = \{00001,\ 00110,\ 11010\}.$$

This is the basic local ambiguity studied throughout the paper: isolated windows can fold together even though Section 4 will show that overlap removes the ambiguity once $m \geq 3$.

Theorem 3.4 (Discrete Fourier formula for the fiber cardinalities)**.** *Let* $\zeta := e^{2\pi i/F_{m+2}}$. *For every* $x \in X_m$ *with* $r_x = \mathrm{Val}_m(x)$,

$$d_m(x) = \frac{1}{F_{m+2}} \sum_{j=0}^{F_{m+2}-1} \zeta^{-jr_x} \prod_{k=1}^{m} \left(1 + \zeta^{jF_{k+1}}\right). \tag{1}$$

Consequently,

$$\sum_{x \in X_m} d_m(x)^2 = \frac{1}{F_{m+2}} \sum_{j=0}^{F_{m+2}-1} \left| \prod_{k=1}^{m} \left(1 + \zeta^{jF_{k+1}}\right) \right|^2. \tag{2}$$

Proof. By Theorem 3.3,

$$d_m(x) = \sum_{\omega \in \{0,1\}^m} \mathbf{1}_{\{\mathrm{Val}_m(\omega) \equiv r_x \pmod{F_{m+2}}\}}.$$

Use the finite Fourier inversion formula on $\mathbb{Z}/F_{m+2}\mathbb{Z}$:

$$\mathbf{1}_{\{n \equiv r_x\}} = \frac{1}{F_{m+2}} \sum_{j=0}^{F_{m+2}-1} \zeta^{j(n-r_x)}.$$

Therefore

$$d_m(x) = \frac{1}{F_{m+2}} \sum_{j=0}^{F_{m+2}-1} \zeta^{-jr_x} \sum_{\omega \in \{0,1\}^m} \zeta^{j\mathrm{Val}_m(\omega)}.$$

Since

$$\mathrm{Val}_m(\omega) = \sum_{k=1}^{m} \omega_k F_{k+1},$$

the inner sum factorizes:

$$\sum_{\omega \in \{0,1\}^m} \zeta^{j\mathrm{Val}_m(\omega)} = \prod_{k=1}^{m} \sum_{\omega_k \in \{0,1\}} \zeta^{j\omega_k F_{k+1}} = \prod_{k=1}^{m} (1 + \zeta^{jF_{k+1}}).$$

This proves (1).

For (2), square (1), sum over $x \in X_m$, and use that r_x runs through the complete residue system $\{0, \ldots, F_{m+2} - 1\}$ by Proposition 3.1. Orthogonality of characters collapses the double sum. □

Remark 3.4. Theorem 3.4 makes $d_m(x)$ an explicit arithmetic object: fiber cardinalities enter the entropy decomposition of Section 5 and the uniform collision statistics of Proposition 7.2 quantitatively. Figure 1 displays the resulting fiber-size distributions for several values of m.

3.4 Affine transfer to a shifted-weight residue model

For auxiliary arithmetic arguments it is convenient to compare the canonical valuation

$$\mathrm{Val}_m(\omega) = \sum_{j=1}^{m} \omega_j F_{j+1}$$

with the shifted-weight residue model

$$\widetilde{\mathrm{Val}}_m(\omega) := \sum_{j=1}^{m} \omega_j F_j \qquad \text{in } \mathbb{Z}/F_{m+2}\mathbb{Z}.$$

By Proposition 3.1, every residue $r \in \{0, \dots, F_{m+2} - 1\}$ has a unique canonical representative

$$x_r := \Big(\mathrm{Val}_m|_{X_m}\Big)^{-1}(r) \in X_m.$$

Accordingly we write

$$d_m^\sharp(r) := d_m(x_r)$$

for the canonical fiber size indexed by residue, and

$$\widetilde{d}_m(r) := \#\Big\{\omega \in \{0,1\}^m : \widetilde{\mathrm{Val}}_m(\omega) \equiv r \pmod{F_{m+2}}\Big\}$$

for the shifted-weight fiber count.

Lemma 3.5 (Fibonacci cross-multiplication modulo F_{m+2})**.** *Extend the Fibonacci sequence by $F_0 := 0$. Then for every $m \geq 1$ and $1 \leq j \leq m$,*

$$F_{m+1}F_j \equiv (-1)^{j-1} F_{m+2-j} \pmod{F_{m+2}}.$$

Proof. Use d'Ocagne's identity

$$F_u F_{v+1} - F_{u+1} F_v = (-1)^v F_{u-v}$$

with $(u, v) = (m+1, j-1)$. This gives

$$F_{m+1}F_j - F_{m+2}F_{j-1} = (-1)^{j-1} F_{m+2-j}.$$

Reducing modulo F_{m+2} proves the claim. □

Theorem 3.6 (Affine transfer principle)**.** *For every $m \geq 1$, define*

$$C_m := \sum_{\substack{1 \leq k \leq m \\ k \not\equiv m \pmod 2}} F_{k+1}$$

and the affine residue map

$$\pi_m(r) := F_{m+1} r + C_m \pmod{F_{m+2}}.$$

Then π_m is a bijection of $\mathbb{Z}/F_{m+2}\mathbb{Z}$, and

$$\widetilde{d}_m(r) = d_m^\sharp\Big(\pi_m(r)\Big) \qquad \text{for all } r \in \mathbb{Z}/F_{m+2}\mathbb{Z}.$$

In particular, the canonical and shifted-weight models have identical fiber multisets.

Proof. Define a map $\tau_m : \{0,1\}^m \to \{0,1\}^m$ by

$$\tau_m(\omega)_k := \begin{cases} \omega_{m+1-k}, & k \equiv m \pmod 2, \\ 1-\omega_{m+1-k}, & k \not\equiv m \pmod 2. \end{cases}$$

It is a composition of bit reversal with parity-selective bit flips, hence a bijection.

We now compute $\mathrm{Val}_m(\tau_m(\omega))$ modulo F_{m+2}. By the definition of τ_m,

$$\mathrm{Val}_m(\tau_m(\omega)) = \sum_{\substack{1\le k\le m \\ k\equiv m \pmod 2}} \omega_{m+1-k}F_{k+1} + \sum_{\substack{1\le k\le m \\ k\not\equiv m \pmod 2}} (1-\omega_{m+1-k})F_{k+1}.$$

Separating the constant part gives

$$\mathrm{Val}_m(\tau_m(\omega)) = \sum_{k=1}^{m}(-1)^{m-k}\omega_{m+1-k}F_{k+1} + C_m.$$

Re-index with $j = m+1-k$:

$$\mathrm{Val}_m(\tau_m(\omega)) = \sum_{j=1}^{m}(-1)^{j-1}\omega_j F_{m+2-j} + C_m.$$

Applying Lemma 3.5 termwise to $\widetilde{\mathrm{Val}}_m(\omega) = \sum_{j=1}^{m}\omega_j F_j$ yields

$$F_{m+1}\widetilde{\mathrm{Val}}_m(\omega) \equiv \sum_{j=1}^{m}(-1)^{j-1}\omega_j F_{m+2-j} \pmod{F_{m+2}}.$$

Comparing with the previous display gives the intertwining identity

$$\mathrm{Val}_m(\tau_m(\omega)) \equiv F_{m+1}\widetilde{\mathrm{Val}}_m(\omega) + C_m = \pi_m\Big(\widetilde{\mathrm{Val}}_m(\omega)\Big) \pmod{F_{m+2}}.$$

Since consecutive Fibonacci numbers are coprime, $\gcd(F_{m+1}, F_{m+2}) = 1$, so multiplication by F_{m+1} is invertible modulo F_{m+2} and π_m is a bijection. Therefore, for every residue r,

$$\begin{aligned} \tilde{d}_m(r) &= \#\Big\{\omega : \widetilde{\mathrm{Val}}_m(\omega) \equiv r \pmod{F_{m+2}}\Big\} \\ &= \#\Big\{\omega : \mathrm{Val}_m(\tau_m(\omega)) \equiv \pi_m(r) \pmod{F_{m+2}}\Big\} \\ &= \#\Big\{u : \mathrm{Val}_m(u) \equiv \pi_m(r) \pmod{F_{m+2}}\Big\} \\ &= d^{\sharp}_m\Big(\pi_m(r)\Big), \end{aligned}$$

where the third equality uses that τ_m is a bijection and the last equality is Theorem 3.3 written in residue coordinates. This proves the theorem. □

Corollary 3.7 (Transfer of multiset-based statistics)**.** *For every $m \ge 1$, the shifted-weight model and the canonical model have the same multiset of fiber sizes. In particular they have the same:*

(i) maximal fiber size,

(ii) maximal-fiber multiplicity,

(iii) second moment $\sum_x d_m(x)^2$,

(iv) every statistic that depends only on the unordered fiber multiset.

Proof. Theorem 3.6 identifies the two residue-count functions up to the permutation π_m. □

Remark 3.5 (Closed form of the affine offset). The parity sum in Theorem 3.6 simplifies to

$$C_m = F_{m+1} - 1.$$

Indeed, for $m = 2q$ one has

$$C_{2q} = F_2 + F_4 + \cdots + F_{2q} = F_{2q+1} - 1 = F_{m+1} - 1,$$

and for $m = 2q + 1$ one has

$$C_{2q+1} = F_3 + F_5 + \cdots + F_{2q+1} = F_{2q+2} - 1 = F_{m+1} - 1.$$

Thus the affine transfer may also be written as

$$\pi_m(r) = F_{m+1}(r + 1) - 1 \pmod{F_{m+2}}.$$

Proposition 3.8 (Affine reflection symmetry of the canonical fiber spectrum)**.** *Let*

$$B_m := \sum_{k=1}^{m} F_{k+1} = F_{m+3} - 2.$$

Then for every residue $r \in \mathbb{Z}/F_{m+2}\mathbb{Z}$,

$$d_m^\sharp(r) = d_m^\sharp(B_m - r).$$

Equivalently, the canonical residue-indexed fiber spectrum is invariant under the affine reflection

$$r \longmapsto B_m - r \qquad \text{in } \mathbb{Z}/F_{m+2}\mathbb{Z}.$$

Proof. Consider the bitwise complement involution

$$\iota : \{0,1\}^m \to \{0,1\}^m, \qquad \iota(\omega)_k := 1 - \omega_k.$$

It is a bijection, and

$$\mathrm{Val}_m(\iota(\omega)) = \sum_{k=1}^{m}(1 - \omega_k)F_{k+1} = \sum_{k=1}^{m} F_{k+1} - \mathrm{Val}_m(\omega) \equiv B_m - \mathrm{Val}_m(\omega) \pmod{F_{m+2}}.$$

Therefore ι sends the preimage of residue r bijectively onto the preimage of residue $B_m - r$. By the definition of $d_m^\sharp$, this proves

$$d_m^\sharp(r) = d_m^\sharp(B_m - r).$$

□

Corollary 3.9 (Symmetry of the uniform visible law)**.** *Let μ_m^{unif} be the uniform law on $\{0,1\}^m$ and let*

$$\pi_m^{\mathrm{unif}} := (\mathrm{Fold}_m)_* \mu_m^{\mathrm{unif}}$$

be the induced visible law on X_m. Then

$$\pi_m^{\mathrm{unif}}(x_r) = \pi_m^{\mathrm{unif}}(x_{B_m - r}) \qquad \text{for every } r \in \mathbb{Z}/F_{m+2}\mathbb{Z}.$$

In particular, every odd centered residue moment of the uniform visible law vanishes.

Proof. Directly from the definition of the pushforward,

$$\pi_m^{\mathrm{unif}}(x_r) = \frac{d_m^\sharp(r)}{2^m}.$$

The claim now follows from Proposition 3.8. □

A related truncation-versus-stabilization effect can also be studied at finite range, but it is not needed here.

4 Overlap graph, sharp threshold, and branch locus

The local residue fibers from Section 3 become a dynamical question only after one imposes overlap. The theorem chain in this section is the structural core of the paper: Theorem 4.1 gives the bounded-degree sofic presentation, Lemma 4.2 isolates the threshold arithmetic at $m = 3$, Theorem 4.3 proves injectivity for every $m \geq 3$, Theorem 4.6 classifies the unique exceptional branch at $m = 2$, and Theorem 4.12 upgrades the above-threshold picture to finite-memory conjugacy and finite-type closure. For background on sofic shifts and labeled graph presentations we refer to [17, 13, 14].

Definition 4.1 (Sliding stabilized-window process)**.** Define

$$\Phi_m : \{0,1\}^{\mathbb{Z}} \longrightarrow (X_m)^{\mathbb{Z}}$$

by

$$(\Phi_m(a))_t := \mathrm{Fold}_m(a_t a_{t+1} \cdots a_{t+m-1}), \qquad t \in \mathbb{Z}.$$

Let

$$Y_m := \Phi_m(\{0,1\}^{\mathbb{Z}}) \subset (X_m)^{\mathbb{Z}}.$$

Definition 4.2 (The overlap graph)**.** Let $\mathcal{G}_m$ be the directed labeled graph defined as follows: the vertex set is

$$V_m := \{0,1\}^{m-1},$$

and for each $u = (u_1, \ldots, u_{m-1}) \in V_m$ and each $c \in \{0,1\}$ there is an edge

$$u \longrightarrow v = (u_2, \ldots, u_{m-1}, c)$$

with label

$$\ell(u,c) := \mathrm{Fold}_m(u_1, \ldots, u_{m-1}, c) \in X_m.$$

Theorem 4.1 (Finite-state lift and bounded multiplicity)**.** *For every $m \geq 2$:*

(i) The shift Y_m is exactly the label shift presented by $\mathcal{G}_m$. In particular, Y_m is a sofic shift.

(ii) The labeled graph $\mathcal{G}_m$ is resolving in both time directions: for each fixed vertex u, the two outgoing edges labeled by $c = 0$ and $c = 1$ have distinct labels, and for each fixed terminal vertex the two incoming edges also have distinct labels.

(iii) The map Φ_m is finite-to-one, with degree at most 2^{m-1}. More precisely, once the initial overlap state

$$(a_0, \ldots, a_{m-2}) \in \{0,1\}^{m-1}$$

is fixed, the entire raw sequence $a \in \{0,1\}^{\mathbb{Z}}$ is uniquely determined by its stabilized-window image $\Phi_m(a)$.

Proof. (i) Let $a \in \{0,1\}^{\mathbb{Z}}$ and define vertices

$$u_t := (a_t, \dots, a_{t+m-2}) \in \{0,1\}^{m-1}.$$

Then

$$u_t \to u_{t+1}$$

is an edge in $\mathcal{G}_m$ labeled by

$$\mathrm{Fold}_m(a_t, \dots, a_{t+m-1}) = (\Phi_m(a))_t.$$

Thus every point of Y_m is realized as the label sequence of a bi-infinite path in $\mathcal{G}_m$.

Conversely, given any bi-infinite path $(u_t)_{t\in\mathbb{Z}}$ in $\mathcal{G}_m$, the overlap condition

$$u_{t+1} = (u_{t,2}, \dots, u_{t,m-1}, c_t)$$

defines a unique bi-infinite binary sequence a whose length-$(m-1)$ windows are the u_t. The corresponding label sequence is exactly $\Phi_m(a)$. Hence Y_m is the label shift of $\mathcal{G}_m$.

(ii) Fix $u \in V_m$. The two outgoing edges correspond to the length-m words $u0$ and $u1$. Their Fibonacci values differ by exactly F_{m+1}:

$$\mathrm{Val}_m(u1) - \mathrm{Val}_m(u0) = F_{m+1}.$$

Since $0 < F_{m+1} < F_{m+2}$, the two residues modulo F_{m+2} are distinct. By Proposition 3.1, distinct residues correspond to distinct labels in X_m. Therefore the two outgoing labels are distinct.

Now fix a terminal vertex $v = (v_1, \dots, v_{m-1}) \in V_m$. Its two incoming edges come from the predecessor vertices

$$(0, v_1, \dots, v_{m-2}) \qquad \text{and} \qquad (1, v_1, \dots, v_{m-2}),$$

both ending at v with appended bit v_{m-1}. The corresponding length-m words differ only in the first coordinate, so their Fibonacci values differ by exactly $F_2 = 1$. Hence their residues modulo F_{m+2} are distinct, and the two incoming labels are distinct as well.

(iii) By (ii), once an initial vertex $u_0 \in V_m$ and a label sequence $y = (y_t)_{t\in\mathbb{Z}} \in Y_m$ are given, there is at most one outgoing edge from the current vertex with the prescribed label y_t. This determines the forward path $(u_t)_{t\geq 0}$. The same item gives at most one incoming edge with prescribed label y_{t-1} into a given terminal vertex u_t, so the path is uniquely determined backward as well. Hence the full bi-infinite path, and therefore the raw sequence a, are uniquely determined by u_0 and y. Since there are exactly $|V_m| = 2^{m-1}$ possible initial vertices, the fiber cardinality of Φ_m over any $y \in Y_m$ is at most 2^{m-1}. □

The arithmetic content of the threshold is already visible at $m = 3$. Combining the three output residues r_0, r_1, r_2 isolates the sum $a_0 + a_1$ together with the tail bit a_4; once a_4 is known, the remaining bits are recovered from the injective maps $(u, v) \mapsto u + 2v$ on $\{0,1\}^2$. Figure 2 summarizes this reconstruction.

Lemma 4.2 (Explicit three-window decoder)**.** *The map*

$$\Phi_{3,3} : \{0,1\}^5 \longrightarrow X_3^3$$

is injective.

Table 1: Diagnostic residues for the exceptional branch $q = 2$ in Lemma 4.2.

(a_0, a_1, a_4)	possible r_0	if $r_0 = 3$, possible r_2
$(1, 1, 0)$	$\{1, 3\}$	$\{0, 2\}$
$(0, 0, 1)$	$\{0, 3\}$	$\{1, 4\}$

Proof. Write the raw block as $a_0a_1a_2a_3a_4$ and let

$$M := F_5 = 5, \qquad r_t := \mathrm{Val}_3\Big(\mathrm{Fold}_3(a_ta_{t+1}a_{t+2})\Big) \in \{0, 1, 2, 3, 4\}, \qquad 0 \le t \le 2.$$

Because $\mathrm{Val}_3|_{X_3}$ is a bijection, the stabilized output triple is equivalent to the residue triple (r_0, r_1, r_2).

Using

$$r_t \equiv a_t + 2a_{t+1} + 3a_{t+2} \pmod 5,$$

set

$$q := r_0 - r_1 - r_2 \pmod 5.$$

The Fibonacci recurrence gives

$$q \equiv a_0 + a_1 + 2a_4 \pmod 5.$$

Since $a_0 + a_1 \in \{0, 1, 2\}$ and $2a_4 \in \{0, 2\}$, one has:

$$q \in \{0, 1\} \Longrightarrow a_4 = 0, \qquad q \in \{3, 4\} \Longrightarrow a_4 = 1.$$

Only the case $q = 2$ requires an extra branch distinction. Then the two possible triples (a_0, a_1, a_4) are

$$(1, 1, 0) \qquad \text{or} \qquad (0, 0, 1).$$

The corresponding diagnostic residues are summarized in Table 1.

Indeed, in the first branch one has

$$r_0 \equiv 3 + 3a_2 \pmod 5, \qquad r_2 \equiv a_2 + 2a_3 \pmod 5,$$

whereas in the second branch

$$r_0 \equiv 3a_2 \pmod 5, \qquad r_2 \equiv a_2 + 2a_3 + 3 \pmod 5.$$

Hence (q, r_0, r_2) determines a_4 uniquely.

Once a_4 is known, define

$$t := r_2 - 3a_4 \pmod 5.$$

Then

$$t \equiv a_2 + 2a_3 \pmod 5.$$

Because the map

$$\{0, 1\}^2 \to \{0, 1, 2, 3\}, \qquad (u, v) \mapsto u + 2v$$

is injective, (a_2, a_3) is uniquely determined by t. Finally define

$$u := r_0 - 3a_2 \pmod 5.$$

Then

$$u \equiv a_0 + 2a_1 \pmod 5,$$

and the same injectivity recovers (a_0, a_1). Thus $a_0a_1a_2a_3a_4$ is uniquely determined by (r_0, r_1, r_2), proving injectivity. □

Theorem 4.3 (Sharp three-window threshold)**.** *Let $m \geq 3$. For each $n \geq m$, define*

$$\Phi_{m,n} : \{0,1\}^{n+m-1} \longrightarrow X_m^n$$

by

$$\Phi_{m,n}(a_0 \dots a_{n+m-2}) := \Big(\mathrm{Fold}_m(a_t \dots a_{t+m-1})\Big)_{t=0}^{n-1}.$$

Then $\Phi_{m,n}$ is injective.

Proof. If $m = 3$, first consider $n = 3$. This is exactly Lemma 4.2. If $n > 3$, the first three stabilized outputs determine the initial raw block

$$a_0 a_1 a_2 a_3 a_4$$

uniquely by Lemma 4.2. In particular the initial overlap state

$$(a_0, a_1) \in \{0,1\}^2$$

is recovered. The right-resolving property in Theorem 4.1(ii) then recursively determines each subsequent raw bit from the current overlap state and the next stabilized symbol. Hence $\Phi_{3,n}$ is injective for every $n \geq 3$.

Assume from now on that $m \geq 4$.

Fix $a_0, \dots, a_{n+m-2} \in \{0,1\}$ and write

$$M := F_{m+2}, \qquad r_t := \mathrm{Val}_m\Big(\mathrm{Fold}_m(a_t \dots a_{t+m-1})\Big) \in \{0, \dots, M-1\}.$$

By Proposition 3.1, the residue block $(r_t)_{t=0}^{n-1}$ is equivalent to the stabilized output block $\Phi_{m,n}(a_0 \dots a_{n+m-2})$.

For $0 \leq t \leq m-3$, define

$$q_t := r_t - r_{t+1} - r_{t+2} \pmod{M}.$$

Using

$$r_t = \sum_{j=0}^{m-1} a_{t+j} F_{j+2},$$

the Fibonacci recurrence gives

$$q_t \equiv a_t + a_{t+1} + F_m a_{t+m+1} \pmod{M}.$$

Indeed, the coefficients of a_{t+j} cancel for $2 \leq j \leq m-1$, the coefficient of a_{t+m} is

$$-F_{m+1} - F_m = -F_{m+2} \equiv 0 \pmod{M},$$

and the coefficient of a_{t+m+1} is

$$-F_{m+1} \equiv F_m \pmod{M}.$$

Since $m \geq 4$, one has $F_m \geq 3$. Hence the sets

$$\{0,1,2\} \qquad \text{and} \qquad \{F_m, F_m+1, F_m+2\}$$

are disjoint in $\{0, \dots, M-1\}$. Therefore each q_t uniquely determines both

$$a_{t+m+1} \in \{0,1\} \qquad \text{and} \qquad s_t := a_t + a_{t+1} \in \{0,1,2\}.$$

From $(q_0, \ldots, q_{m-3})$ we recover all tail bits

$$a_{m+1}, a_{m+2}, \ldots, a_{2m-2},$$

and all adjacent prefix sums

$$s_0 = a_0 + a_1,\ s_1 = a_1 + a_2,\ \ldots,\ s_{m-3} = a_{m-3} + a_{m-2}.$$

Next, use the last two residues among the first m windows. Define

$$u := r_{m-1} - \sum_{j=2}^{m-1} a_{m-1+j} F_{j+2} \pmod{M}.$$

All terms except a_{m-1} and a_m have been removed, so

$$u \equiv a_{m-1} F_2 + a_m F_3 = a_{m-1} + 2a_m \pmod{M}.$$

Because $a_{m-1} + 2a_m \in \{0, 1, 2, 3\} \subset [0, M-1]$, this congruence is an ordinary integer identity, and (a_{m-1}, a_m) is uniquely determined.

Similarly, define

$$v := r_{m-2} - \sum_{j=1}^{m-1} a_{m-2+j} F_{j+2} \pmod{M}.$$

Then

$$v \equiv a_{m-2} F_2 = a_{m-2} \pmod{M},$$

so a_{m-2} is uniquely determined.

Now recover the remaining prefix bits by backward recursion:

$$a_{m-3} = s_{m-3} - a_{m-2}, \quad a_{m-4} = s_{m-4} - a_{m-3}, \quad \ldots, \quad a_0 = s_0 - a_1.$$

Thus the first m stabilized outputs already determine the entire raw block

$$a_0, \ldots, a_{2m-2}.$$

If $n = m$, the proof is complete. If $n > m$, the recovered initial overlap state

$$(a_0, \ldots, a_{m-2}) \in \{0, 1\}^{m-1}$$

together with Theorem 4.1(ii) implies that each subsequent label uniquely determines the next raw bit. Iterating this right-resolving decoding recovers

$$a_{2m-1}, \ldots, a_{n+m-2}.$$

Therefore $\Phi_{m,n}$ is injective. □

Corollary 4.4 (Explicit sliding-block decoder above threshold)**.** *Assume $m \geq 3$, and let*

$$\Psi_m : \mathcal{L}_m(Y_m) \to \{0, 1\}^{2m-1}$$

be the inverse of $\Phi_{m,m}$ onto its image. Define

$$\psi_m(w) := (\Psi_m(w))_1 \qquad \text{for } w \in \mathcal{L}_m(Y_m).$$

Then for every $a \in \{0, 1\}^{\mathbb{Z}}$ and $y = \Phi_m(a)$,

$$a_t = \psi_m(y_t, \ldots, y_{t+m-1}) \qquad \text{for all } t \in \mathbb{Z}.$$

Thus the inverse factor map has memory 0 and anticipation $m - 1$.

Proof. For each $t \in \mathbb{Z}$, the block

$$w_t := (y_t, \ldots, y_{t+m-1})$$

lies in $\mathcal{L}_m(Y_m)$ and is the image under $\Phi_{m,m}$ of the raw block

$$(a_t, \ldots, a_{t+2m-2}).$$

By Theorem 4.3, $\Phi_{m,m}$ is injective, so

$$\Psi_m(w_t) = (a_t, \ldots, a_{t+2m-2}).$$

Taking the first coordinate gives

$$\psi_m(w_t) = a_t,$$

as required. □

Proposition 4.5 (Sharp low-window counterexamples)**.** *The threshold in Theorem 4.3 is sharp.*

(i) At $m = 2$, sequence-level injectivity fails:

$$\Phi_2(0^{\mathbb{Z}}) = \Phi_2(1^{\mathbb{Z}}) = (\ldots, 00, 00, 00, \ldots).$$

(ii) At $m = 3$, the requirement $n \geq m$ is sharp:

$$\Phi_{3,2}(0001) = \Phi_{3,2}(0110) = (000, 001).$$

Proof. For (i), $\mathrm{Fold}_2(00) = 00$ and

$$\mathrm{Val}_2(11) = F_2 + F_3 = 3 \equiv 0 \pmod{F_4},$$

so $\mathrm{Fold}_2(11) = 00$ as well. Hence every length-2 window in both constant sequences stabilizes to 00.

For (ii), one has

$$\mathrm{Fold}_3(000) = 000, \qquad \mathrm{Fold}_3(001) = 001,$$

and also

$$\mathrm{Val}_3(011) = F_3 + F_4 = 5 \equiv 0 \pmod{F_5}, \qquad \mathrm{Val}_3(110) = F_2 + F_3 = 3,$$

so

$$\mathrm{Fold}_3(011) = 000, \qquad \mathrm{Fold}_3(110) = 001.$$

Thus the two raw blocks 0001 and 0110 have the same stabilized output pair. □

Theorem 4.6 (Complete classification of the two-window branch locus)**.** *For $n \geq 1$, define*

$$\Phi_{2,n} : \{0,1\}^{n+1} \longrightarrow X_2^n, \qquad \Phi_{2,n}(a_0 \ldots a_n) := \Big(\mathrm{Fold}_2(a_t a_{t+1})\Big)_{t=0}^{n-1},$$

and write

$$\mathcal{Y}_{2,n} := \Phi_{2,n}(\{0,1\}^{n+1}).$$

Then

$$\Phi_{2,n}^{-1}\Big((00)^n\Big) = \{0^{n+1}, 1^{n+1}\},$$

and for every $y \in \mathcal{Y}_{2,n} \setminus \{(00)^n\}$ one has

$$|\Phi_{2,n}^{-1}(y)| = 1.$$

On two-sided path space,

$$\Phi_2^{-1}\Big((\ldots, 00, 00, \ldots)\Big) = \{0^{\mathbb{Z}}, 1^{\mathbb{Z}}\},$$

and every other point of Y_2 has singleton fiber.

Proof. At $m = 2$ one has

$$\mathrm{Fold}_2(00) = 00, \qquad \mathrm{Fold}_2(11) = 00, \qquad \mathrm{Fold}_2(01) = 01, \qquad \mathrm{Fold}_2(10) = 10.$$

Thus the stabilized symbol 00 records equality of adjacent raw bits, while 01 and 10 record the two possible local jumps.

Fix $y = (y_0, \ldots, y_{n-1}) \in \mathcal{Y}_{2,n}$. If $y = (00)^n$, then every adjacent pair in the raw block must consist of equal bits, so the only possibilities are 0^{n+1} and 1^{n+1}. Hence

$$\Phi_{2,n}^{-1}\Big((00)^n\Big) = \{0^{n+1}, 1^{n+1}\}.$$

If $y \neq (00)^n$, choose t with $y_t \in \{01, 10\}$. Then y_t determines the pair (a_t, a_{t+1}) exactly. For $s = t+1, \ldots, n-1$, the symbol y_s determines a_{s+1} from a_s: if $y_s = 00$ then $a_{s+1} = a_s$, if $y_s = 01$ then $(a_s, a_{s+1}) = (0, 1)$, and if $y_s = 10$ then $(a_s, a_{s+1}) = (1, 0)$. The same rule propagates uniquely backward for $s = t-1, \ldots, 0$. Therefore the entire raw block is uniquely determined, so every other fiber is a singleton.

The bi-infinite statement is proved in the same way. If $y = (\ldots, 00, 00, \ldots)$, then every adjacent raw pair is constant, so the only preimages are $0^{\mathbb{Z}}$ and $1^{\mathbb{Z}}$. Otherwise choose any t with $y_t \in \{01, 10\}$ and propagate the reconstruction uniquely in both time directions. □

Corollary 4.7 (Exact entropy and KL defect formulas at the two-window threshold)**.** *Fix $n \geq 1$, let $A \sim P$ be a law on $\{0,1\}^{n+1}$, and set $Y := \Phi_{2,n}(A)$. Write*

$$p_0 := P(0^{n+1}), \qquad p_1 := P(1^{n+1}), \qquad q := p_0 + p_1,$$

and define the binary entropy

$$h(r) := -r \log r - (1-r)\log(1-r), \qquad 0 \leq r \leq 1.$$

Then

$$H_P(A \mid Y) = q\, h\Big(\frac{p_0}{q}\Big) \leq q \log 2,$$

with the convention that the right-hand side is 0 when $q = 0$.

If Q is another law on $\{0,1\}^{n+1}$, with

$$q_0 := Q(0^{n+1}), \qquad q_1 := Q(1^{n+1}), \qquad s := q_0 + q_1,$$

then the KL defect vanishes when $q = 0$. If $q > 0$ and $s > 0$, one has

$$\begin{aligned} D_{\mathrm{KL}}(P\|Q) - D_{\mathrm{KL}}\Big((\Phi_{2,n})_* P \| (\Phi_{2,n})_* Q\Big) \\ = q\left[\frac{p_0}{q}\log\frac{p_0/q}{q_0/s} + \frac{p_1}{q}\log\frac{p_1/q}{q_1/s}\right]. \end{aligned} \tag{3}$$

Here terms with $p_i = 0$ are interpreted as 0.

Proof. By Theorem 4.6, every conditional fiber of $\Phi_{2,n}$ is a singleton except the branch fiber over $(00)^n$, which is exactly $\{0^{n+1}, 1^{n+1}\}$. Therefore $H_P(A \mid Y = y) = 0$ for $y \neq (00)^n$, while on the branch fiber the conditional law is

$$P(A = 0^{n+1} \mid Y = (00)^n) = \frac{p_0}{q}, \qquad P(A = 1^{n+1} \mid Y = (00)^n) = \frac{p_1}{q}.$$

Hence

$$H_P(A \mid Y) = P(Y = (00)^n)\, h\Big(\frac{p_0}{q}\Big) = q\, h\Big(\frac{p_0}{q}\Big),$$

and the bound follows from $h(r) \leq \log 2$.

For the KL identity, all singleton fibers cancel when one compares the raw divergence with the pushed-forward divergence. Only the branch fiber contributes:

$$\begin{aligned} D_{\mathrm{KL}}(P\|Q) - D_{\mathrm{KL}}\Big((\Phi_{2,n})_* P \| (\Phi_{2,n})_* Q\Big) \\ = p_0 \log \frac{p_0}{q_0} + p_1 \log \frac{p_1}{q_1} - q \log \frac{q}{s}. \end{aligned}$$

If $q = 0$ this is 0. Otherwise $s > 0$ by assumption, so factor out q and rewrite the remaining expression in terms of the two-point distributions on the branch fiber, which gives (3). □

Corollary 4.8 (Asymptotic sufficiency off the branch fiber)**.** *Let $\{P_{u,n} : u \in \mathcal{U}\}$ be any family of laws on $\{0,1\}^{n+1}$, and suppose that*

$$\sup_{u \in \mathcal{U}} P_{u,n}\Big(\{0^{n+1}, 1^{n+1}\}\Big) \leq \varepsilon_n.$$

Then

$$\sup_{u \in \mathcal{U}} H_{P_{u,n}}\Big(A \mid \Phi_{2,n}(A)\Big) \leq \varepsilon_n \log 2.$$

In particular, if ε_n decays exponentially, then two-window stabilized readout is exponentially asymptotically sufficient for this family.

Proof. Apply Corollary 4.7 and use $h(r) \leq \log 2$. □

Corollary 4.9 (Mod-zero invertibility at the two-window threshold)**.** *Let ν be a shift-invariant Borel probability measure on $\{0,1\}^{\mathbb{Z}}$ such that*

$$\nu(\{0^{\mathbb{Z}}, 1^{\mathbb{Z}}\}) = 0,$$

and set $\eta := (\Phi_2)_* \nu$. Then

$$\Phi_2 : \Big(\{0,1\}^{\mathbb{Z}}, \nu\Big) \longrightarrow (Y_2, \eta)$$

is a measure-theoretic isomorphism.

Proof. By Theorem 4.6, the restriction of Φ_2 to

$$\{0,1\}^{\mathbb{Z}} \setminus \{0^{\mathbb{Z}}, 1^{\mathbb{Z}}\}$$

is one-to-one, with image

$$Y_2 \setminus \{(\ldots, 00, 00, \ldots)\}.$$

Since Φ_2 is continuous, this restricted map is Borel, and its inverse on the image is Borel by the standard Lusin–Souslin theorem for injective Borel maps between Polish spaces. Both the branch set and its image have measure zero under ν and η, respectively. Hence the restriction of Φ_2 furnishes an invertible, shift-equivariant correspondence modulo null sets, which is exactly a measure-theoretic isomorphism. □

Corollary 4.10 (Full block complexity of the one-sided stabilized image)**.** *Let*

$$p_{Y_m^+}(n) := \#\mathcal{L}_n(Y_m^+)$$

denote the length-n word complexity of the one-sided stabilized image

$$Y_m^+ := \Phi_m^+(\{0,1\}^{\mathbb{N}}), \qquad (\Phi_m^+(a))_t := \mathrm{Fold}_m(a_t \dots a_{t+m-1}).$$

If $m \geq 3$ and $n \geq m$, then

$$p_{Y_m^+}(n) = 2^{n+m-1}.$$

Proof. The domain of $\Phi_{m,n}$ has cardinality 2^{n+m-1}. By Theorem 4.3, $\Phi_{m,n}$ is injective, so its image has the same cardinality. But that image is precisely the set of length-n blocks appearing in Y_m^+. □

Corollary 4.11 (One-sided stabilized-window conjugacy)**.** *If $m \geq 3$, the one-sided sliding stabilized-window map*

$$\Phi_m^+ : \left(\{0,1\}^{\mathbb{N}}, \sigma\right) \longrightarrow (Y_m^+, \sigma)$$

is a topological conjugacy.

Proof. If $\Phi_m^+(a) = \Phi_m^+(b)$ for $a, b \in \{0,1\}^{\mathbb{N}}$, then for every $n \geq m$ the two sequences have the same image under $\Phi_{m,n}$. By Theorem 4.3, their first $n+m-1$ raw bits coincide. Letting $n \to \infty$ shows $a = b$. Thus Φ_m^+ is injective.

It is continuous, shift-equivariant, and surjective onto Y_m^+ by definition. Since $\{0,1\}^{\mathbb{N}}$ is compact and Y_m^+ is Hausdorff, a continuous bijection onto Y_m^+ is a homeomorphism. Hence Φ_m^+ is a topological conjugacy. □

Theorem 4.12 (Two-sided finite-memory conjugacy and finite-type closure)**.** *Assume $m \geq 3$. Then:*

(i) The two-sided stabilized-window map

$$\Phi_m : \left(\{0,1\}^{\mathbb{Z}}, \sigma\right) \to (Y_m, \sigma)$$

is injective, hence a topological conjugacy onto its image.

(ii) There exists a local decoder

$$\psi_m : \mathcal{L}_m(Y_m) \to \{0,1\}$$

such that for every $a \in \{0,1\}^{\mathbb{Z}}$ and $y = \Phi_m(a)$,

$$a_t = \psi_m(y_t, \dots, y_{t+m-1}) \qquad \text{for all } t \in \mathbb{Z}.$$

Thus the inverse conjugacy is a sliding-block code with memory 0 and anticipation $m-1$.

(iii) The image shift Y_m is a subshift of finite type of memory at most m.

Proof. (ii) is exactly Corollary 4.4.

If $\Phi_m(a) = \Phi_m(b)$, then for every t the corresponding length-m output blocks agree, so by (ii) one has

$$a_t = \psi_m(y_t, \dots, y_{t+m-1}) = b_t.$$

Hence $a = b$, proving injectivity. Since Φ_m is continuous, shift-equivariant, and surjective onto Y_m by definition, (i) follows.

For (iii), let $y \in (X_m)^{\mathbb{Z}}$. For each t, if the length-m block

$$w_t := (y_t, \dots, y_{t+m-1})$$

belongs to $\mathcal{L}_m(Y_m)$, define

$$B_t := \Psi_m(w_t) \in \{0,1\}^{2m-1}.$$

If $y = \Phi_m(a)$ for some a, then these decoded blocks must satisfy the overlap consistency condition

$$(B_t)_2(B_t)_3 \cdots (B_t)_{2m-1} = (B_{t+1})_1(B_{t+1})_2 \cdots (B_{t+1})_{2m-2} \qquad \text{for all } t \in \mathbb{Z}. \tag{4}$$

Conversely, suppose that every length-m block of y lies in $\mathcal{L}_m(Y_m)$ and that the decoded blocks satisfy (4) for all t. Then the blocks (B_t) glue uniquely to a bi-infinite raw sequence $a \in \{0,1\}^{\mathbb{Z}}$, and by construction every length-m output block of $\Phi_m(a)$ agrees with the corresponding block of y. Hence $\Phi_m(a) = y$.

The conditions "$w_t \in \mathcal{L}_m(Y_m)$" and (4) depend only on the length-$(m+1)$ block $(y_t, \dots, y_{t+m})$. Since there are only finitely many such blocks, they define a finite forbidden list. Therefore Y_m is a subshift of finite type of memory at most m. □

Remark 4.1 (Decoder complexity). The inverse code ψ_m of Theorem 4.12(ii) reads a sliding window of length m and outputs one raw bit. Equivalently, one may implement the decoder by maintaining an overlap state of size $m-1$. The arithmetic cost is $O(m)$ per output symbol, since each decoding step involves at most m Fibonacci-weighted additions modulo F_{m+2}.

Proposition 4.13 (Immediate dynamical corollaries above threshold)**.** *Assume $m \geq 3$.*

(i) For every $n \geq 1$,

$$\#\operatorname{Fix}(\sigma^n|_{Y_m}) = 2^n, \qquad \zeta_{Y_m}(z) = \frac{1}{1-2z}.$$

(ii) Pushforward by Φ_m induces a bijection between the shift-invariant Borel probability measures on $\{0,1\}^{\mathbb{Z}}$ and those on Y_m.

(iii) Ergodicity, weak mixing, mixing, Bernoulli-ness, K-property, and measure-theoretic entropy are transported exactly through the conjugacy.

Proof. All three items are immediate from Theorem 4.12. Fixed-point counts and the Artin–Mazur zeta function are conjugacy invariants. The inverse sliding-block code gives the bijection of invariant measures. The dynamical statements in (iii) are standard invariants of topological or measure-theoretic conjugacy; see [29]. The thermodynamic consequences for continuous potentials are recorded separately in Theorem 6.4. □

Remark 4.2. For $m \geq 3$ the stabilization image is not merely a sofic factor of bounded degree. It is a subshift of finite type topologically conjugate to the full two-shift by a finite-memory code, and the only obstruction below threshold is the explicit branch locus of Theorem 4.6.

Remark 4.3. In the terminology of [17], $\mathcal{G}_m$ is a right-resolving presentation of the sofic shift Y_m, and Φ_m has degree at most 2^{m-1} (Theorem 4.1(iii)). A right-resolving factor of degree 1 between irreducible shifts of finite type is a conjugacy [17, Theorem 8.1.16]; see also [36] for the equal-entropy resolving theory and [37] for the lower-entropy obstruction. The content of the threshold theorem is that degree 1 is achieved at $m = 3$ and not before; this is an arithmetic fact that cannot be read off from the general sofic framework and requires the explicit decoder of Lemma 4.2.

Proposition 4.14 (Measure-theoretic entropy preservation under the stabilized-window factor)**.** *Let ν be a shift-invariant Borel probability measure on $\{0,1\}^{\mathbb{Z}}$, and let*

$$\eta := (\Phi_m)_*\nu$$

be its image on $Y_m \subset (X_m)^{\mathbb{Z}}$. Then

$$h_\eta(\sigma) = h_\nu(\sigma).$$

Proof. For $n \geq 1$, let

$$A_{0:n+m-2} := (A_0, \dots, A_{n+m-2})$$

denote the raw binary block of length $n + m - 1$, and let

$$Y_{0:n-1} := (Y_0, \dots, Y_{n-1}), \qquad Y_t = \mathrm{Fold}_m(A_t \cdots A_{t+m-1}),$$

denote the stabilized block of length n.

Since $Y_{0:n-1}$ is a deterministic function of $A_{0:n+m-2}$, one has

$$H_\eta(Y_{0:n-1}) \leq H_\nu(A_{0:n+m-2}).$$

Conversely, by Theorem 4.1(iii), once the initial overlap state

$$(A_0, \dots, A_{m-2}) \in \{0,1\}^{m-1}$$

is specified, the entire raw block $A_{0:n+m-2}$ is uniquely determined by the stabilized block $Y_{0:n-1}$. Therefore

$$H_\nu(A_{0:n+m-2} \mid Y_{0:n-1}) \leq (m-1)\log 2.$$

By the chain rule,

$$H_\nu(A_{0:n+m-2}) = H_\eta(Y_{0:n-1}) + H_\nu(A_{0:n+m-2} \mid Y_{0:n-1}),$$

hence

$$0 \leq H_\nu(A_{0:n+m-2}) - H_\eta(Y_{0:n-1}) \leq (m-1)\log 2.$$

Divide by n and let $n \to \infty$. The additive discrepancy is $O(1)$, so it disappears after normalization. The left-hand side converges to $h_\nu(\sigma)$ because shifting the block length by the fixed amount $m - 1$ does not affect the entropy rate, while the right-hand side converges to $h_\eta(\sigma)$. Thus

$$h_\eta(\sigma) = h_\nu(\sigma).$$

□

Corollary 4.15 (Sequence-level entropy preservation)**.** *For every $m \geq 2$,*

$$h_{\mathrm{top}}(Y_m) = \log 2.$$

Proof. The map Φ_m is a continuous factor map from the full two-shift onto Y_m, so

$$h_{\mathrm{top}}(Y_m) \le h_{\mathrm{top}}(\{0,1\}^{\mathbb{Z}}) = \log 2.$$

$$\text{Let } \nu_{1/2} := \left(\tfrac{1}{2}, \tfrac{1}{2}\right)^{\otimes \mathbb{Z}}$$

be the Bernoulli measure on the full two-shift, and let

$$\eta_{1/2} := (\Phi_m)_* \nu_{1/2}.$$

By Proposition 4.14,

$$h_{\eta_{1/2}}(\sigma) = h_{\nu_{1/2}}(\sigma) = \log 2.$$

Since $\eta_{1/2}$ is a shift-invariant Borel probability measure on Y_m, the variational principle [29] gives

$$h_{\mathrm{top}}(Y_m) \ge \log 2.$$

Therefore equality holds. □

Remark 4.4. For every $m \ge 2$ the stabilized-window factor preserves entropy. Once $m \ge 3$, the picture is much more rigid: Y_m is conjugate to the full two-shift, so periodic data, invariant measures, pressure, and equilibrium states are transported exactly. Proposition 4.5 shows that this threshold is sharp.

5 Local aliasing on isolated windows

The previous section shows that overlap removes local ambiguity above threshold. The purpose of this section is therefore narrower: it quantifies the aliasing created by isolated windows before overlap is imposed. The resulting entropy/KL identity is intentionally local and should not be confused with a path-space irreversibility statement.

5.1 Fiber-uniform lift

Definition 5.1 (Fiber-uniform lift)**.** For a distribution $\pi \in \Delta(X_m)$ on stabilized types, the *fiber-uniform lift* is the probability measure $\mu^e \in \Delta(\{0,1\}^m)$ defined by

$$\mu^e(\omega) := \frac{\pi(\mathrm{Fold}_m(\omega))}{|\mathcal{F}_m(\mathrm{Fold}_m(\omega))|}, \qquad \omega \in \{0,1\}^m.$$

Remark 5.1. The normalization is immediate:

$$\sum_{\omega \in \{0,1\}^m} \mu^e(\omega) = \sum_{x \in X_m} \sum_{\omega \in \mathcal{F}_m(x)} \frac{\pi(x)}{|\mathcal{F}_m(x)|} = \sum_{x \in X_m} \pi(x) = 1.$$

5.2 Exact entropy ledger

Theorem 5.1 (Entropy decomposition over fibers)**.** *Let $\mu \in \Delta(\{0,1\}^m)$ be any distribution on raw words, $\pi := (\mathrm{Fold}_m)_*\mu \in \Delta(X_m)$ its pushforward under stabilization, and μ^e the fiber-uniform lift of π. Write $d_m(x) := |\mathcal{F}_m(x)|$. Then*

$$H(\mu) = H(\pi) + \mathbb{E}_\pi[\log d_m(X)] - D_{\mathrm{KL}}(\mu \| \mu^e), \tag{5}$$

where $H(\cdot)$ denotes Shannon entropy and D_{KL} the Kullback–Leibler divergence.

Proof. By definition,

$$D_{\mathrm{KL}}(\mu\|\mu^e) = -H(\mu) - \sum_{\omega\in\{0,1\}^m} \mu(\omega)\log\mu^e(\omega).$$

For $\omega \in \mathcal{F}_m(x)$ one has $\log\mu^e(\omega) = \log\pi(x) - \log d_m(x)$. Grouping the sum by fibers gives

$$\sum_{\omega} \mu(\omega)\log\pi(\mathrm{Fold}_m(\omega)) = \sum_{x\in X_m} \pi(x)\log\pi(x) = -H(\pi),$$

and

$$\sum_{\omega} \mu(\omega)\log d_m(\mathrm{Fold}_m(\omega)) = \sum_{x\in X_m} \pi(x)\log d_m(x) = \mathbb{E}_\pi[\log d_m(X)].$$

Substituting these identities and rearranging yields (5). □

Corollary 5.2 (Entropy increase under fiber equilibration)**.** *Let μ^e be the fiber-uniform lift of* $(\mathrm{Fold}_m)_*\mu$*. Then*

$$H(\mu^e) - H(\mu) = D_{\mathrm{KL}}(\mu\|\mu^e). \tag{6}$$

Proof. Directly from Definition 5.1,

$$H(\mu^e) = H(\pi) + \mathbb{E}_\pi[\log d_m(X)].$$

Subtracting (5) gives (6). □

5.3 Consequences for general and benchmark microstate laws

Remark 5.2. Identity (5) separates three contributions: the visible entropy $H(\pi)$ of the stabilized types, the logarithmic fiber degeneracy $\mathbb{E}_\pi[\log d_m(X)]$, and the within-fiber disequilibrium $D_{\mathrm{KL}}(\mu\|\mu^e)$. For uniform input on $\{0,1\}^m$, one has $\pi_m^{\mathrm{unif}}(x) = d_m(x)/2^m$, so the fiber counts determine both the visible law and the collision probability discussed in Section 7.

Remark 5.3. Theorem 5.1 is intentionally a fixed-window identity. It quantifies the local aliasing of isolated words. Section 6 shows that above threshold this defect does not survive on path space, because overlap reconstructs the raw trajectory by a finite-memory conjugacy.

6 Finite-memory conjugacy and dynamical consequences

Theorem 4.12 turns the sharp threshold into a dynamical statement. Above threshold, stabilization is not merely a bounded-degree sofic factor; it is a finite-memory conjugacy onto a finite-type image. This section records the resulting consequences for block presentations, path-space equivalence, and thermodynamic formalism.

6.1 Block presentations above the threshold

The theorem below is best viewed as a deterministic equivalence between raw and stabilized block presentations. The language of sufficiency is a corollary of this bijection, not the primary structural input.

Fix $m \geq 3$ and $n \geq m$. Write

$$\mathcal{Y}_{m,n} := \Phi_{m,n}\Big(\{0,1\}^{n+m-1}\Big) \subset X_m^n$$

for the legal length-n stabilized output blocks, and let

$$\Psi_{m,n} : \mathcal{Y}_{m,n} \to \{0,1\}^{n+m-1}$$

denote the inverse of $\Phi_{m,n}$ from Theorem 4.3.

Theorem 6.1 (Deterministic equivalence of block presentations above threshold)**.** *Fix $m \geq 3$ and $n \geq m$. Let $\{P_u : u \in \mathcal{U}\}$ be any family of probability laws on the raw block space $\{0,1\}^{n+m-1}$, and let*

$$Q_u := (\Phi_{m,n})_* P_u$$

be the induced family on $\mathcal{Y}_{m,n}$. Then:

(i) For every $u \in \mathcal{U}$, if $A^{(n)} \sim P_u$ and $Y^{(n)} := \Phi_{m,n}(A^{(n)})$, then

$$A^{(n)} = \Psi_{m,n}(Y^{(n)}) \qquad P_u\text{-a.s.}$$

Consequently,

$$H_{P_u}(A^{(n)} \mid Y^{(n)}) = 0.$$

(ii) For any random parameter U with conditional law $A^{(n)} \mid \{U = u\} \sim P_u$,

$$I(U; Y^{(n)}) = I(U; A^{(n)}).$$

(iii) The raw-block and stabilized-block experiments are deterministically equivalent. In particular, for every $u, v \in \mathcal{U}$,

$$D_{\mathrm{KL}}(P_u \| P_v) = D_{\mathrm{KL}}(Q_u \| Q_v), \qquad D_{\mathrm{TV}}(P_u, P_v) = D_{\mathrm{TV}}(Q_u, Q_v),$$

the Chernoff information agrees, and every Bayesian or Neyman–Pearson decision problem has the same optimal risk under the two experiments.

Proof. Theorem 4.3 states that $\Phi_{m,n}$ is injective, so $\Psi_{m,n}$ is well-defined on $\mathcal{Y}_{m,n}$ and satisfies

$$\Psi_{m,n}(\Phi_{m,n}(a)) = a \qquad \text{for every } a \in \{0,1\}^{n+m-1}.$$

Hence $A^{(n)} = \Psi_{m,n}(Y^{(n)})$ almost surely under every P_u, proving (i).

For (ii), $Y^{(n)}$ is a deterministic function of $A^{(n)}$, so by data processing

$$I(U; Y^{(n)}) \leq I(U; A^{(n)}).$$

Conversely, $A^{(n)}$ is a deterministic function of $Y^{(n)}$ by (i), so the reverse inequality also holds.

For (iii), each Q_u is the pushforward of P_u under a bijection between $\{0,1\}^{n+m-1}$ and $\mathcal{Y}_{m,n}$. Therefore any divergence or decision criterion that is invariant under deterministic bijections is identical for the two experiments. The displayed identities for D_{KL} and D_{TV} are obtained by rewriting the defining sums on $\mathcal{Y}_{m,n}$ via the inverse map $\Psi_{m,n}$. Chernoff information is preserved for the same reason, and a decision rule on one experiment transports to the other by composition with $\Phi_{m,n}$ or $\Psi_{m,n}$ without changing its loss distribution. □

6.2 Path-space equivalence and the localization of irreversibility

The point of this subsection is to separate the fixed-window aliasing ledger from any claim about process-level loss. Above threshold the stabilized process is measurably isomorphic to the raw binary process, so divergence rates are preserved exactly. The many-to-one character of Fold_m on isolated windows therefore records only local aliasing once overlap is taken into account.

Corollary 6.2 (Path-space equivalence and divergence-rate preservation)**.** *Assume $m \geq 3$, let $\{\nu_u : u \in \mathcal{U}\}$ be a family of shift-invariant Borel probability measures on $\{0,1\}^{\mathbb{Z}}$, and write*

$$\eta_u := (\Phi_m)_* \nu_u$$

for their images on Y_m. Then Φ_m is a measurable isomorphism between the path-space experiments $(\{0,1\}^{\mathbb{Z}}, \nu_u)$ and (Y_m, η_u). In particular, for every $u, v \in \mathcal{U}$, the finite-block Kullback–Leibler divergences agree exactly after the block identification by $\Phi_{m,n}$, so the relative entropy rates satisfy

$$h(\nu_u \| \nu_v) = h(\eta_u \| \eta_v)$$

whenever these specific relative entropy rates exist. If $\nu_v = \nu_u^{\mathrm{rev}}$ is the time-reversal of ν_u and the entropy-production rate is defined by $h(\nu_u \| \nu_u^{\mathrm{rev}})$, then the same identity holds for entropy-production rates.

Proof. The path-space isomorphism is exactly Theorem 4.12(i)–(ii). For every $n \geq m$, the block map $\Phi_{m,n}$ identifies the length-$(n+m-1)$ raw marginals of ν_u and ν_v with the length-n stabilized marginals of η_u and η_v. The finite-block Kullback–Leibler divergences therefore coincide term by term. Dividing by n and letting $n \to \infty$ yields the equality of specific relative entropies whenever these limits exist. The entropy-production statement is the special case in which the reference law is the time reversal of the forward process and the rate is defined through the corresponding specific relative entropy. □

Proposition 6.3 (Localization of irreversibility above threshold)**.** *Assume $m \geq 3$. Let ν be a shift-invariant Borel probability measure on $\{0,1\}^{\mathbb{Z}}$, let $\eta := (\Phi_m)_* \nu$, and let μ_m and π_m denote the corresponding length-m marginals on $\{0,1\}^m$ and X_m. Then:*

(i) the finite-window entropy ledger of Theorem 5.1,

$$H(\mu_m) = H(\pi_m) + \mathbb{E}_{\pi_m}[\log d_m(X)] - D_{\mathrm{KL}}(\mu_m \| \mu_m^e),$$

describes the unresolved multiplicity of isolated windows;

(ii) the path-space experiments $(\{0,1\}^{\mathbb{Z}}, \nu)$ and (Y_m, η) are measurably isomorphic, so specific relative entropy and, when defined through time reversal, entropy-production rate are the same on both sides.

Consequently canonical admissibilization above threshold does not generate a new extensive source of irreversibility on path space.

Proof. Item (i) is exactly Theorem 5.1, which is formulated at fixed window length m. Item (ii) is the single-process specialization of Corollary 6.2. Since the path-space laws are measurably isomorphic, any extensive information loss would have to appear in the specific relative entropy or entropy-production rate; item (ii) excludes this. Therefore the degeneracy term from item (i) is a local aliasing effect of isolated windows rather than a process-level source of irreversibility. □

6.3 Exact transport of continuous potentials and finite-range energetics

This subsection records the main dynamical and thermodynamic consequences of the finite-memory conjugacy. By transport we mean only passage through the finite-memory conjugacy. No renormalization-group flow, fixed point, or universality claim is intended. The first result applies to arbitrary continuous potentials; the finite-range formulas that follow give the explicit local form relevant for local observables and benchmark Hamiltonians.

Theorem 6.4 (Transport of continuous potentials under the conjugacy)**.** *Assume $m \geq 3$, let $\varphi : \{0,1\}^{\mathbb{Z}} \to \mathbb{R}$ be continuous, and define*

$$\widetilde{\varphi} := \varphi \circ \Phi_m^{-1} : Y_m \to \mathbb{R}.$$

Then:

(i) for every shift-invariant Borel probability measure μ on $\{0,1\}^{\mathbb{Z}}$,

$$\int_{Y_m} \widetilde{\varphi}\, d(\Phi_{m*}\mu) = \int \varphi\, d\mu, \qquad h_{\Phi_{m*}\mu}(\sigma) = h_\mu(\sigma);$$

(ii) the topological pressures agree:

$$P_{Y_m}(\widetilde{\varphi}) = P_{\{0,1\}^{\mathbb{Z}}}(\varphi);$$

(iii) μ is an equilibrium state for φ if and only if $\Phi_{m*}\mu$ is an equilibrium state for $\widetilde{\varphi}$.

Proof. Theorem 4.12(i) states that Φ_m is a topological conjugacy onto Y_m, so Φ_m^{-1} is continuous and $\widetilde{\varphi}$ is again continuous. The integral identity is immediate from the definition of pushforward. The entropy identity is Proposition 4.14. Therefore for every invariant measure μ,

$$h_{\Phi_{m*}\mu}(\sigma) + \int \widetilde{\varphi}\, d(\Phi_{m*}\mu) = h_\mu(\sigma) + \int \varphi\, d\mu.$$

Taking the supremum over invariant measures and using the variational principle on both systems yields the pressure identity. The equilibrium-state correspondence is then immediate from equality of the variational functional under pushforward. □

Corollary 6.5 (Parameterized phase diagrams are transported exactly)**.** *Assume $m \geq 3$, and let $\{\varphi_\lambda : \lambda \in \Lambda\}$ be any family of continuous potentials on the raw shift. Then for every parameter value λ,*

$$P_{Y_m}(\widetilde{\varphi}_\lambda) = P_{\{0,1\}^{\mathbb{Z}}}(\varphi_\lambda), \qquad \widetilde{\varphi}_\lambda := \varphi_\lambda \circ \Phi_m^{-1}.$$

Hence every phase boundary or nonanalytic point defined through the pressure function, and every equilibrium-state correspondence defined by the variational principle, is preserved pointwise under stabilized readout.

Proof. Apply Theorem 6.4 to each parameter value separately. □

Proposition 6.6 (Finite-range energetics under local inversion)**.** *Assume $m \geq 3$. Let $\phi : \{0,1\}^r \to \mathbb{R}$ be a finite-range interaction of range $r \geq 1$ on the raw shift. Define the induced interaction on Y_m by*

$$\widetilde{\phi}(y_t, \dots, y_{t+r+m-2}) := \phi\Big(\psi_m(y_t, \dots, y_{t+m-1}), \dots, \psi_m(y_{t+r-1}, \dots, y_{t+r+m-2})\Big).$$

Then $\widetilde{\phi}$ has range $r + m - 1$ on Y_m, and for every $a \in \{0,1\}^{\mathbb{Z}}$, $y := \Phi_m(a)$, and $n \geq 1$,

$$\sum_{t=0}^{n-1} \phi(a_t, \dots, a_{t+r-1}) = \sum_{t=0}^{n-1} \widetilde{\phi}(y_t, \dots, y_{t+r+m-2}).$$

Proof. For every $t \in \mathbb{Z}$, Theorem 4.12(ii) gives

$$a_t = \psi_m(y_t, \dots, y_{t+m-1}).$$

Substituting this identity into $\phi(a_t, \dots, a_{t+r-1})$ yields the definition of $\widetilde{\phi}$ and proves the equality of the finite-volume energy sums term by term. □

Corollary 6.7 (Thermodynamic transport above threshold)**.** *Assume $m \geq 3$. For every finite-range interaction ϕ on the raw full shift, the induced interaction $\widetilde{\phi}$ on Y_m satisfies:*

(i) finite-volume Hamiltonians agree exactly on matched bi-infinite configurations and therefore differ by at most an $O(1)$ boundary term under any standard finite-volume boundary convention;

(ii) the corresponding partition functions agree up to boundary factors independent of volume, so pressure and free-energy density coincide;

(iii) equilibrium states are transported bijectively by Φ_m.

In this restricted sense the stabilized readout transports finite-range interactions exactly above the three-window threshold.

Proof. Item (i) is Proposition 6.6. Item (ii) follows because the energy sums agree exactly on matched configurations, and any discrepancy between finite-volume conventions comes only from the fixed-width boundary needed to evaluate the range-$(r + m - 1)$ potential on Y_m. The pressure and free-energy-density statements are obtained by dividing the logarithms of the partition functions by volume and letting the volume tend to infinity. Item (iii) is Proposition 4.13(iii) applied to the potential $\widetilde{\phi}$ transported through the conjugacy. □

Corollary 6.8 (Occupancy and nearest-neighbour energy under stabilized readout)**.** *Assume $m \geq 3$ and interpret $A_t \in \{0,1\}$ as a binary occupancy readout along the scan. Define the finite-volume observables*

$$N_n(A) := \sum_{t=0}^{n-1} A_t, \qquad C_n(A) := \sum_{t=0}^{n-1} A_t A_{t+1},$$

and the nearest-neighbour energy

$$\mathcal{H}_n(A) := -J\, C_n(A) - h\, N_n(A).$$

Then for $Y = \Phi_m(A)$ there exist local observables on the stabilized process,

$$\widetilde{A}_t(Y) := \psi_m(Y_t, \dots, Y_{t+m-1}), \qquad \widetilde{C}_t(Y) := \widetilde{A}_t(Y)\widetilde{A}_{t+1}(Y),$$

such that

$$N_n(A) = \sum_{t=0}^{n-1} \widetilde{A}_t(Y), \qquad C_n(A) = \sum_{t=0}^{n-1} \widetilde{C}_t(Y), \qquad \mathcal{H}_n(A) = \widetilde{\mathcal{H}}_n(Y)$$

with

$$\widetilde{\mathcal{H}}_n(Y) := -J \sum_{t=0}^{n-1} \widetilde{C}_t(Y) - h \sum_{t=0}^{n-1} \widetilde{A}_t(Y).$$

Consequently, occupancy density, pair density, and the corresponding free-energy density are preserved exactly by stabilized readout.

Proof. The identities for N_n and C_n follow immediately from Theorem 4.12(ii), which recovers each occupancy bit as a local observable of the stabilized trajectory. The energy identity is the special case of Proposition 6.6 with range $r = 2$. □

6.4 Secondary information-geometric consequences

This subsection is logically downstream from the conjugacy and thermodynamic transport results. Above threshold the block experiments are related by a literal bijection, not merely by a sufficient statistic, so the same deterministic equivalence also preserves the block-level information geometry.

Corollary 6.9 (Exact transport of block-level information geometry)**.** *Fix $m \geq 3$ and $n \geq m$.*

(i) For any convex function $f : [0, \infty) \to \mathbb{R}$ with $f(1) = 0$, let

$$D_f(P\|Q) := \sum_a Q(a)\, f\left(\frac{P(a)}{Q(a)}\right)$$

be the associated Csiszar f-divergence on the finite raw block space, with the usual conventions at $Q(a) = 0$. Then for any probability laws P, Q on $\{0,1\}^{n+m-1}$,

$$D_f(P\|Q) = D_f\Big((\Phi_{m,n})_* P \| (\Phi_{m,n})_* Q\Big).$$

In particular, the Hellinger integrals and the Bhattacharyya coefficient are preserved exactly.

(ii) Let $\{P_\theta : \theta \in \Theta\}$ be a family of strictly positive mass functions on $\{0,1\}^{n+m-1}$ such that $p_\theta(a) := P_\theta(\{a\})$ is C^1 in θ for every raw block a. Let $Q_\theta := (\Phi_{m,n})_* P_\theta$ and write

$$q_\theta(y) := Q_\theta(\{y\}), \qquad y \in \mathcal{Y}_{m,n}.$$

Then

$$q_\theta(y) = p_\theta(\Psi_{m,n}(y)), \qquad \nabla_\theta \log q_\theta(y) = \nabla_\theta \log p_\theta(\Psi_{m,n}(y)).$$

Consequently the Fisher information matrices of the raw and stabilized block experiments are identical:

$$\mathcal{I}_Y(\theta) = \mathcal{I}_A(\theta).$$

Proof. For (i), $\Phi_{m,n}$ is a bijection between the raw block space and $\mathcal{Y}_{m,n}$, with inverse $\Psi_{m,n}$. Writing the defining sum for the f-divergence on $\mathcal{Y}_{m,n}$ and changing variables by $a = \Psi_{m,n}(y)$ gives the identity immediately. The Hellinger and Bhattacharyya quantities are special cases obtained by choosing the corresponding functions of the likelihood ratio.

For (ii), the mass-function identity

$$q_\theta(y) = p_\theta(\Psi_{m,n}(y))$$

is exactly the pushforward relation under a bijection. Differentiating with respect to θ gives the score identity. The Fisher information matrix is the expectation of the score outer product, so the same change of variables under $\Psi_{m,n}$ shows that the raw and stabilized expectations coincide. □

7 Arithmetic fallout: a second-moment recurrence

The threshold and conjugacy theorems do not require a detailed study of the entire fiber spectrum, but one unconditional statistic is useful later and illustrates the remaining local arithmetic: the second moment

$$\Sigma_m := \sum_{x \in X_m} d_m(x)^2,$$

equivalently the collision probability of the visible law under uniform raw input.

7.1 Second moment and collision probability

Theorem 7.1 (Second-moment linear recurrence)**.** *For all $m \geq 4$,*

$$\Sigma_m = 2\,\Sigma_{m-1} + 2\,\Sigma_{m-2} - 2\,\Sigma_{m-3}, \tag{7}$$

with $\Sigma_1 = 2$, $\Sigma_2 = 6$, $\Sigma_3 = 14$.

Proof. The second moment counts raw pairs sharing a residue modulo F_{m+2}. Write $\epsilon_k := \omega_k - \omega'_k \in \{-1, 0, 1\}$ with multiplicity weights $w(\pm 1) = 1$, $w(0) = 2$, and define

$$f(n, t) := \text{weighted count of } \epsilon \in \{-1, 0, 1\}^n \text{ with } \sum_{k=1}^{n} \epsilon_k F_{k+1} = t.$$

Since

$$\left| \sum_{k=1}^{m} \epsilon_k F_{k+1} \right| \leq \sum_{k=1}^{m} F_{k+1} = F_{m+3} - 2 < 2F_{m+2},$$

only the values $0, \pm F_{m+2}$ contribute, so by symmetry

$$\Sigma_m = f(m, 0) + 2f(m, F_{m+2}). \tag{8}$$

Conditioning on the last digit gives

$$f(m, F_{m+2}) = 2f(m-1, F_{m+2}) + f(m-1, F_m) + f(m-1, F_{m+3}),$$

and the first and third terms vanish by the same range bound. Hence

$$f(m, F_{m+2}) = f(m-1, F_m).$$

Conditioning once more at level $m-1$ yields

$$f(m-1, F_m) = 2f(m-2, F_m) + f(m-2, 0),$$

because the term at $2F_m$ is out of range for $m \geq 4$. Using (8) at window $m-2$, this is exactly Σ_{m-2}.

Likewise,

$$f(m, 0) = 2f(m-1, 0) + 2f(m-1, F_{m+1}) = f(m-1, 0) + \Sigma_{m-1}.$$

Writing $a_m := f(m, 0)$, one has $a_m = \Sigma_m - 2\Sigma_{m-2}$ and $a_{m-1} = \Sigma_{m-1} - 2\Sigma_{m-3}$. Eliminating a_m gives (7). The initial values are obtained by direct enumeration. □

Proposition 7.2 (Uniform visible law and collision probability)**.** *Let μ_m^{unif} be the uniform distribution on $\{0,1\}^m$, and let $\pi_m^{\text{unif}} := (\text{Fold}_m)_* \mu_m^{\text{unif}}$ be the induced distribution on X_m. Then for every $x \in X_m$,*

$$\pi_m^{\text{unif}}(x) = \frac{d_m(x)}{2^m},$$

and therefore

$$\text{Col}_{\text{unif}}(m) := \sum_{x \in X_m} \pi_m^{\text{unif}}(x)^2 = \frac{1}{2^{2m}} \sum_{x \in X_m} d_m(x)^2. \tag{9}$$

Proof. For $x \in X_m$, the fiber $\mathcal{F}_m(x)$ has cardinality $d_m(x)$ by definition. Since μ_m^{unif} assigns mass 2^{-m} to each raw word,

$$\pi_m^{\text{unif}}(x) = \mu_m^{\text{unif}}(\mathcal{F}_m(x)) = \frac{d_m(x)}{2^m}.$$

Substituting into $\text{Col}_{\text{unif}}(m) = \sum_x \pi_m^{\text{unif}}(x)^2$ gives (9). □

8 Quantitative consequences in the Sturmian benchmark

Once the finite-memory conjugacy is in place, the benchmark rotation model supplies deterministic quantitative bounds. The point of this section is not to re-establish the threshold theorem, but to measure how orbit discrepancy and boundary blur perturb the transported observables in the specific Sturmian source law.

8.1 Stabilized block laws from discrepancy

Fix a stabilization aperture m and a stabilized block length $n \geq 1$. For the orbit segment $\{x_0 + t\alpha\}_{t=0}^{N-1}$ under the standard two-interval rotation coding, let $\hat{\mu}_{m+n-1,N}$ denote the empirical distribution of the resulting raw words of length $m+n-1$, and let μ_{m+n-1} be the corresponding limiting raw-word distribution given by Lebesgue measure on the depth-$(m+n-1)$ Sturmian atoms. Define the empirical stabilized n-block law and its limit by

$$\hat{\eta}_{m,n,N} := (\Phi_{m,n})_* \hat{\mu}_{m+n-1,N}, \qquad \eta_{m,n} := (\Phi_{m,n})_* \mu_{m+n-1}.$$

For $n = 1$ these are exactly the empirical folded histogram and its limit:

$$\hat{\eta}_{m,1,N} = \hat{\pi}_{m,N}, \qquad \eta_{m,1} = \pi_m^{\text{fold}}.$$

Theorem 8.1 (Stabilized block law from discrepancy)**.** *In the rotation readout model with slope α, initial phase x_0, and empirical stabilized n-block law $\hat{\eta}_{m,n,N}$, one has*

$$D_{\mathrm{TV}}\Big(\hat{\eta}_{m,n,N}, \eta_{m,n}\Big) \le (m+n)\, D_N^*(\alpha; x_0), \tag{10}$$

where $D_N^*(\alpha; x_0)$ is the star discrepancy of the orbit segment $\{x_0 + t\alpha\}_{t=0}^{N-1}$.

Proof. Let $\mathcal{I}_{m+n-1}$ be the partition of $\mathbb{T}^1$ into the nonempty depth-$(m+n-1)$ Sturmian atoms. Standard Sturmian complexity gives $|\mathcal{I}_{m+n-1}| \le m+n$, and each atom $I \in \mathcal{I}_{m+n-1}$ is an interval. The empirical law $\hat{\mu}_{m+n-1,N}$ and the limiting law μ_{m+n-1} are exactly the empirical and Lebesgue masses of these atoms, so

$$D_{\mathrm{TV}}\Big(\hat{\mu}_{m+n-1,N}, \mu_{m+n-1}\Big) = \frac{1}{2} \sum_{I \in \mathcal{I}_{m+n-1}} \Big|\hat{\lambda}_N(I) - \lambda(I)\Big|,$$

where $\hat{\lambda}_N$ is the empirical orbit measure and λ is Lebesgue measure on $\mathbb{T}^1$.

For any interval $I = [a, b) \subset \mathbb{T}^1$,

$$\Big|\hat{\lambda}_N(I) - \lambda(I)\Big| \le \Big|\hat{\lambda}_N([0,b)) - \lambda([0,b))\Big| + \Big|\hat{\lambda}_N([0,a)) - \lambda([0,a))\Big| \le 2D_N^*(\alpha; x_0).$$

Therefore

$$D_{\mathrm{TV}}\Big(\hat{\mu}_{m+n-1,N}, \mu_{m+n-1}\Big) \le \frac{1}{2}|\mathcal{I}_{m+n-1}| \cdot 2D_N^*(\alpha; x_0) \le (m+n) D_N^*(\alpha; x_0).$$

Since $\Phi_{m,n}$ is deterministic, data processing gives

$$D_{\mathrm{TV}}\Big(\hat{\eta}_{m,n,N}, \eta_{m,n}\Big) \le D_{\mathrm{TV}}\Big(\hat{\mu}_{m+n-1,N}, \mu_{m+n-1}\Big),$$

which proves (10). □

Corollary 8.2 (Bound for bounded stabilized-block observables)**.** *Let $G : Y_m^n \to \mathbb{R}$ be any bounded observable. Then*

$$\Big|\sum_{y \in Y_m^n} G(y)\Big(\hat{\eta}_{m,n,N}(y) - \eta_{m,n}(y)\Big)\Big| \le 2\|G\|_\infty D_{\mathrm{TV}}\Big(\hat{\eta}_{m,n,N}, \eta_{m,n}\Big) \le 2(m+n)\|G\|_\infty D_N^*(\alpha; x_0). \tag{11}$$

Proof. For the signed measure

$$\rho := \hat{\eta}_{m,n,N} - \eta_{m,n}$$

on the finite set Y_m^n one has

$$\Big|\sum_{y \in Y_m^n} G(y)\rho(y)\Big| \le \|G\|_\infty \sum_{y \in Y_m^n} |\rho(y)| \le 2\|G\|_\infty D_{\mathrm{TV}}\Big(\hat{\eta}_{m,n,N}, \eta_{m,n}\Big),$$

and Theorem 8.1 gives the second inequality in (11). □

8.2 Boundary-layer stability for stabilized blocks

Let $R_0 : H \to \{0,1\}$ denote the sharp indicator readout and let $R_\rho : H \to \{0,1\}$ be a binary finite-resolution readout obtained from a smoothed kernel with resolution parameter ρ. Write

$$\varepsilon_\rho := \mu\Big(\{h \in H : \ R_\rho(h) \neq R_0(h)\}\Big)$$

for the boundary-layer mass. Let $\mu^{(0)}_{m+n-1}$ and $\mu^{(\rho)}_{m+n-1}$ be the corresponding raw-word marginals on $\{0,1\}^{m+n-1}$, and let

$$\eta^{(0)}_{m,n} := (\Phi_{m,n})_*\mu^{(0)}_{m+n-1}, \qquad \eta^{(\rho)}_{m,n} := (\Phi_{m,n})_*\mu^{(\rho)}_{m+n-1}$$

be the stabilized n-block laws.

Proposition 8.3 (Boundary-layer control for raw and stabilized block laws)**.** *For every $m, n \geq 1$,*

$$D_{\mathrm{TV}}\Big(\mu^{(\rho)}_{m+n-1}, \mu^{(0)}_{m+n-1}\Big) \leq (m+n-1)\varepsilon_\rho, \qquad D_{\mathrm{TV}}\Big(\eta^{(\rho)}_{m,n}, \eta^{(0)}_{m,n}\Big) \leq (m+n-1)\varepsilon_\rho.$$

Proof. Couple the two readout laws by sampling the same source point $h \sim \mu$ and reading the two raw words

$$W^{(0)}_{m+n-1}(h) := \Big(R_0(T^t h)\Big)_{t=0}^{m+n-2}, \qquad W^{(\rho)}_{m+n-1}(h) := \Big(R_\rho(T^t h)\Big)_{t=0}^{m+n-2}.$$

If these two words differ, then for at least one time $t \in \{0, \dots, m+n-2\}$ one has $R_\rho(T^t h) \neq R_0(T^t h)$. By the union bound and T-invariance of μ,

$$\mathbb{P}\Big(W^{(\rho)}_{m+n-1} \neq W^{(0)}_{m+n-1}\Big) \leq \sum_{t=0}^{m+n-2} \mu\Big(\{h : \ R_\rho(T^t h) \neq R_0(T^t h)\}\Big) = (m+n-1)\varepsilon_\rho.$$

The coupling inequality therefore gives the raw-word bound. The stabilized-block bound then follows from data processing under the deterministic map $\Phi_{m,n}$. □

Corollary 8.4 (Resolution/discrepancy certificate)**.** *Assume that the blurred readout R_ρ is still a two-interval coding of the same irrational rotation, so that Theorem 8.1 applies to the empirical stabilized n-block law $\hat{\eta}^{(\rho)}_{m,n,N}$ and its limit $\eta^{(\rho)}_{m,n}$. Then*

$$D_{\mathrm{TV}}\Big(\hat{\eta}^{(\rho)}_{m,n,N}, \eta^{(0)}_{m,n}\Big) \leq (m+n)D^*_N(\alpha; x_0) + (m+n-1)\varepsilon_\rho.$$

Proof. By the triangle inequality,

$$D_{\mathrm{TV}}\Big(\hat{\eta}^{(\rho)}_{m,n,N}, \eta^{(0)}_{m,n}\Big) \leq D_{\mathrm{TV}}\Big(\hat{\eta}^{(\rho)}_{m,n,N}, \eta^{(\rho)}_{m,n}\Big) + D_{\mathrm{TV}}\Big(\eta^{(\rho)}_{m,n}, \eta^{(0)}_{m,n}\Big).$$

The first term is bounded by Theorem 8.1, and the second by Proposition 8.3. □

8.3 Operational recovery of finite-range observables

Theorem 8.5 (Operational recovery of finite-range observables from stabilized readout)**.** *Assume $m \geq 3$ and let $g : \{0,1\}^r \to \mathbb{R}$ be a bounded local observable of range $r \geq 1$. Define the transported observable on stabilized blocks of length $r+m-1$ by*

$$\widetilde{g}(y_t, \dots, y_{t+r+m-2}) := g\Big(\psi_m(y_t, \dots, y_{t+m-1}), \dots, \psi_m(y_{t+r-1}, \dots, y_{t+r+m-2})\Big).$$

Then:

(i) For every matched trajectory $Y = \Phi_m(A)$ and every $t \geq 0$,

$$g(A_t, \dots, A_{t+r-1}) = \widetilde{g}(Y_t, \dots, Y_{t+r+m-2}).$$

Hence every finite-range empirical average on the raw measured process is exactly recoverable from the stabilized trajectory.

(ii) In the sharp irrational-rotation readout,

$$\Big|\frac{1}{N}\sum_{t=0}^{N-1} g(A_t, \dots, A_{t+r-1}) - \sum_{y \in Y_m^{r+m-1}} \widetilde{g}(y)\eta_{m,r+m-1}(y)\Big| \leq 2(r+2m-1)\|g\|_\infty D_N^*(\alpha; x_0).$$

(iii) In the blurred two-interval readout of the same irrational rotation,

$$\Big|\frac{1}{N}\sum_{t=0}^{N-1} g(A_t^{(\rho)}, \dots, A_{t+r-1}^{(\rho)}) - \sum_{y \in Y_m^{r+m-1}} \widetilde{g}(y)\eta_{m,r+m-1}^{(0)}(y)\Big| \leq 2\|g\|_\infty\Big((r+2m-1)D_N^*(\alpha; x_0) + (r+2m-2)\varepsilon_\rho\Big).$$

Proof. Item (i) is a direct substitution of the local inverse from Theorem 4.12(ii). For item (ii), set $n := r+m-1$ and define $G := \widetilde{g}$ on Y_m^n. The empirical average of g along the raw trajectory agrees with the empirical average of G under the stabilized n-block law by item (i), so Corollary 8.2 gives

$$\Big|\frac{1}{N}\sum_{t=0}^{N-1} g(A_t, \dots, A_{t+r-1}) - \sum_{y \in Y_m^n} G(y)\eta_{m,n}(y)\Big| \leq 2(m+n)\|G\|_\infty D_N^*(\alpha; x_0).$$

Since $n = r+m-1$ and $\|G\|_\infty = \|g\|_\infty$, this is exactly the displayed bound in (ii).

For item (iii), item (i) again identifies the raw empirical average with the empirical average of $G = \widetilde{g}$ under the blurred stabilized n-block law. Applying the same bounded-observable estimate to the signed measure $\hat{\eta}_{m,n,N}^{(\rho)} - \eta_{m,n}^{(0)}$ and then using Corollary 8.4 yields

$$\Big|\frac{1}{N}\sum_{t=0}^{N-1} g(A_t^{(\rho)}, \dots, A_{t+r-1}^{(\rho)}) - \sum_{y \in Y_m^n} G(y)\eta_{m,n}^{(0)}(y)\Big| \leq 2\|G\|_\infty\Big((m+n)D_N^*(\alpha; x_0) + (m+n-1)\varepsilon_\rho\Big),$$

which reduces to the stated formula after substituting $n = r+m-1$. □

Corollary 8.6 (Prototype measurement Hamiltonians)**.** *Assume $m \geq 3$, let*

$$N_N(A) := \sum_{t=0}^{N-1} A_t, \qquad C_N(A) := \sum_{t=0}^{N-1} A_t A_{t+1}, \qquad \mathcal{H}_N(A) := -J\, C_N(A) - h\, N_N(A),$$

and let $\bar{N}_m$, $\bar{C}_m$, and $\bar{\mathcal{H}}_m$ denote the corresponding sharp-limit densities transported to the stabilized process. Then:

(i) the identities of Corollary 6.8 recover $N_N(A)$, $C_N(A)$, and $\mathcal{H}_N(A)$ exactly from stabilized readout;

(ii) in the blurred rotation readout one has the explicit finite-sample bounds

$$\left|\frac{1}{N}N_N(A^{(\rho)}) - \bar{N}_m\right| \le 2\Big(2m\, D_N^*(\alpha;x_0) + (2m-1)\varepsilon_\rho\Big),$$

$$\left|\frac{1}{N}C_N(A^{(\rho)}) - \bar{C}_m\right| \le 2\Big((2m+1)\, D_N^*(\alpha;x_0) + 2m\,\varepsilon_\rho\Big),$$

and

$$\left|\frac{1}{N}\mathcal{H}_N(A^{(\rho)}) - \bar{\mathcal{H}}_m\right| \le 2|h|\Big(2m\, D_N^*(\alpha;x_0) + (2m-1)\varepsilon_\rho\Big) + 2|J|\Big((2m+1)\, D_N^*(\alpha;x_0) + 2m\,\varepsilon_\rho\Big).$$

Proof. Apply Theorem 8.5 with $g(a_0) := a_0$ for occupancy and with $g(a_0, a_1) := a_0 a_1$ for pair density. In both cases $\|g\|_\infty = 1$, and the exact identities are already recorded in Corollary 6.8. The energy-density bound follows by combining the occupancy and pair-density estimates linearly with coefficients $-h$ and $-J$. □

Corollary 8.7 (Bounded-type slopes)**.** *Assume that the rotation slope α has bounded partial quotients. Then there exists a constant $C_\alpha > 0$, depending only on the continued-fraction data of α, such that for every bounded local observable $g : \{0,1\}^r \to \mathbb{R}$ and every $N \ge 2$,*

$$\left|\frac{1}{N}\sum_{t=0}^{N-1} g(A_t, \dots, A_{t+r-1}) - \sum_{y\in Y_m^{r+m-1}} \tilde{g}(y)\eta_{m,r+m-1}(y)\right| \le 2(r+2m-1)\|g\|_\infty C_\alpha \frac{\log N}{N}.$$

Proof. For irrational rotations of bounded type, standard discrepancy estimates give

$$D_N^*(\alpha;x_0) \le C_\alpha \frac{\log N}{N}$$

uniformly in the initial phase x_0 [15, 10]. Substituting this into Theorem 8.5(ii) gives the claim. □

Remark 8.1. Theorem 8.5 and Corollary 8.4 quantify the benchmark source law after the conjugacy has already been established. Above threshold, stabilization contributes no new loss mechanism; the only explicit finite-sample error terms are orbit discrepancy and boundary-layer blur.

9 Discussion

The main result is a sharp overlap threshold for a locally many-to-one Fibonacci folding. The word map Fold_m collapses residue classes on isolated windows, but Theorem 4.3 and Theorem 4.12 show that once overlapping windows are imposed, every aperture $m \ge 3$ becomes a finite-memory conjugacy onto a finite-type image.

The failure at $m = 2$ is equally sharp. Rather than a diffuse non-injective regime, one finds a single branch locus: Theorem 4.6 identifies the unique exceptional orbit, Corollary 4.7 computes the exact entropy and Kullback–Leibler defect concentrated there, and Corollary 4.9 shows that outside this branch set the two-window map is already invertible modulo null sets.

These results clarify the status of the entropy ledger. Section 5 quantifies local aliasing of fixed windows, whereas Section 6 proves that above threshold no new process-level irreversibility is created. The finite-memory conjugacy transports invariant measures, pressure, and equilibrium states exactly, so the stabilization image is mathematically a recoding of the full shift rather than a new dynamical system with altered thermodynamic content.

The residue formulas, discrete Fourier representation, second-moment recurrence, and discrepancy bounds then appear in the right role: they are quantitative corollaries of the folding arithmetic and the conjugacy, not substitutes for the main theorem. In the Sturmian benchmark they yield explicit deterministic control of stabilized block laws and of transported local observables.

From the sofic-shift perspective, the threshold theorem can be read as follows: the sliding stabilization map Φ_m is a right-resolving factor of degree at most 2^{m-1} for every $m \geq 2$. General theory guarantees that a right-resolving one-to-one factor between irreducible shifts of finite type is a conjugacy [36, 17]. The non-trivial content is that degree one is achieved at exactly $m = 3$, and that the obstruction at $m = 2$ is confined to a single orbit pair rather than a diffuse set. Neither conclusion follows from the abstract framework; both require the explicit Fibonacci residue arithmetic of Lemma 4.2 and Theorem 4.6.

The Fibonacci setting is used for explicit solvability. It is the smallest aperiodic example in which the threshold decoder, the branch classification, and the thermodynamic transport can all be proved completely. Extending the same picture to more general Ostrowski or model-set numerations would require new base-case decoders and new overlap arithmetic, so that problem remains genuinely open. A natural next case is the family of metallic-mean stabilizations, where the forbidden word changes from 11 to $1 \cdots 1$ of length $k + 1$; the modular arithmetic changes qualitatively and the threshold window is expected to depend on k.

The readout motivation from quasiperiodic order remains useful at the boundary of the paper: it explains why one is naturally led to a local folding defined on the full cube and why finite-sample discrepancy bounds matter. But the theorem-level core established here is a symbolic-dynamical and arithmetic statement about finite-memory recoding.

10 Conclusion

We studied the canonical Fibonacci stabilization map as a local folding on binary words and asked when overlap restores the information lost on isolated windows. The answer is exact: $m = 3$ is the minimal threshold. Above it, sliding stabilized windows determine the raw trajectory by a finite-memory inverse and identify the image with a subshift of finite type conjugate to the full two-shift.

The exceptional window $m = 2$ is fully classified by a unique branch orbit, with exact entropy and Kullback–Leibler defect formulas. This isolates the only genuine obstruction to injectivity and shows that the low-window failure is rigid rather than diffuse.

Once the conjugacy is established, the dynamical consequences are automatic but important: invariant measures, continuous potentials, pressure, and equilibrium states transport exactly. In the Sturmian benchmark the same structure yields explicit discrepancy and boundary-blur bounds for stabilized block laws and finite-range observables.

For the original quasicrystal readout problem, the conclusion is therefore precise. Finite-resolution support inflation makes the full cube unavoidable locally, but above the

sharp three-window threshold the ensuing stabilization is not a lossy coarse graining of trajectories. It is a finite-memory recoding.

A Source-layer model-set input

This appendix records the only cut-and-project facts used in the main text. They justify the benchmark source law and the existence of well-defined patch frequencies; all theorem-level arguments in the body of the paper are symbolic and arithmetic.

Let $E = \mathbb{R}^d$, $H = \mathbb{R}^n$, and let $\Gamma \subset E \times H$ be a lattice such that $\pi_E|_\Gamma$ is injective and $\pi_H(\Gamma)$ is dense. Writing $\Gamma_E := \pi_E(\Gamma)$ and $x^\star := \pi_H(\gamma)$ for the unique $\gamma \in \Gamma$ with $\pi_E(\gamma) = x$, a compact window $W \subset H$ with nonempty interior and Haar-null boundary defines the regular model set

$$\Lambda(W, u) := \{x \in \Gamma_E : \ x^\star \in W + u\} \subset E$$

at phase $u \in H$. Varying u moves along the hull, and the hull has $(E \times H)/\Gamma$ as its natural equicontinuous factor. In particular, a one-dimensional scan is obtained by coding a torus orbit through a finite observation map [5, 7].

Proposition A.1 (Source-layer input from regular model sets)**.** *Let $\Lambda(W, u)$ be a regular model set as above, and let $X_\Lambda(u)$ be its hull in the local topology with the E-translation action. Then $(X_\Lambda(u), E)$ is uniquely ergodic. In particular, for μ_H-a.e. phase u and every finite patch P, the frequency* $\mathrm{freq}(P; \Lambda(W, u))$ *exists and agrees with the unique invariant patch frequency. The scan dynamics can therefore be modeled by a torus translation with a well-defined symbolic factor law [5, 7].*

This is the only place where source-layer cut-and-project input is required. Once that input is fixed, the main argument reduces to the symbolic-arithmetic stabilization theory developed in the body of the paper.

B Finite-range checks

Exact enumeration up to $m = 22$ was used only as a consistency check for the finite-window formulas in the main text. Table 2 lists representative values. Throughout, $F_1 = F_2 = 1$ and $|X_m| = F_{m+2}$. Since the minimum fiber size is always 1, we record only the maximal fiber size and the second moment

$$\Sigma_m := \sum_{x \in X_m} d_m(x)^2.$$

The full enumeration up to $m = 22$ agrees with the growth pattern discussed in the main text, and an independent shifted-weight computation gives the same values, in line with Theorem 3.6.

Data availability statement

No external dataset was used in this study. The finite-range tables and figures can be regenerated from the computational material accompanying the source. All scripts and generated artifacts will be deposited in a public repository upon acceptance.

Table 2: Representative exact fiber data. The second moment Σ_m is the quantity appearing in Theorem 3.4.

m	$\lvert X_m \rvert$	$\max d_m$	Σ_m
3	5	2	14
5	13	4	88
12	377	21	51216
22	46368	233	452899840

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figures_jpa/fig_fiber_distribution.pdf

Figure 1: Fiber-size distributions for the stabilization map Fold_m at window lengths $m = 5, 8, 12$, obtained by exact enumeration over all 2^m binary words. Each bar gives the number of residue classes in X_m whose fiber has a given size $d_m(x)$. The distributions are highly non-uniform. They encode the combinatorial content of the coarse-graining: both the entropy ledger of Theorem 5.1 and the collision probability of Proposition 7.2 are determined by this fiber spectrum.

Step 1. From (r_0, r_1, r_2) compute $q = r_0 - r_1 - r_2 \pmod 5$.
Step 2. If $q \in \{0, 1\}$ then $a_4 = 0$; if $q \in \{3, 4\}$ then $a_4 = 1$; if $q = 2$ use (r_0, r_2) to separate the two branches.
Step 3. Recover (a_2, a_3) from $t = r_2 - 3a_4 \equiv a_2 + 2a_3 \pmod 5$.
Step 4. Recover (a_0, a_1) from $u = r_0 - 3a_2 \equiv a_0 + 2a_1 \pmod 5$.

Figure 2: Schematic of the three-window decoder. The quantity $q = r_0 - r_1 - r_2 \pmod 5$ isolates the tail bit except on a single branch, which is then resolved by (r_0, r_2).